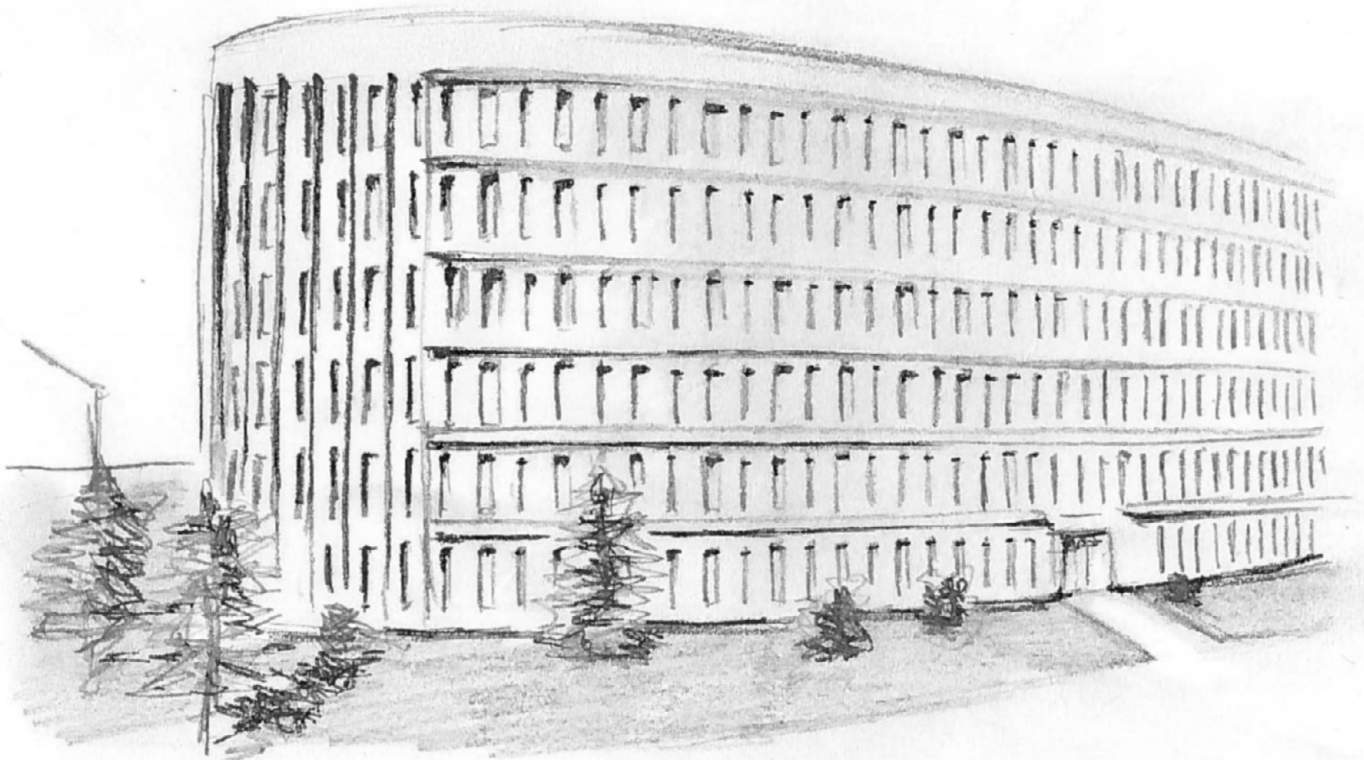
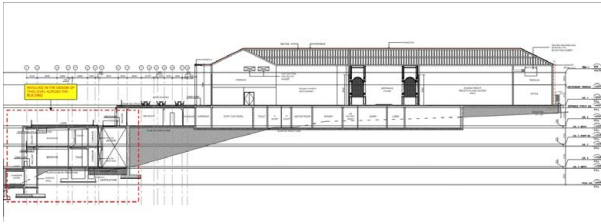


ENGINEERING PORTFOLIO

Anmol Tanwar (GIStructE)



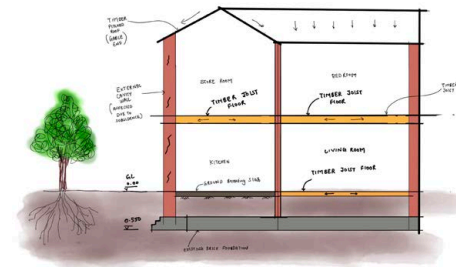
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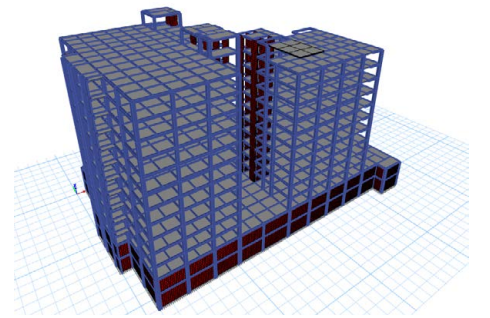
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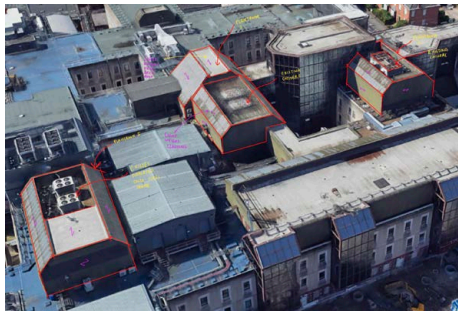
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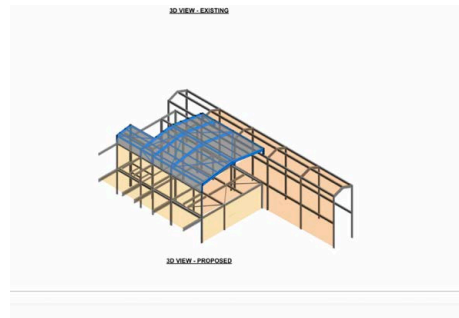
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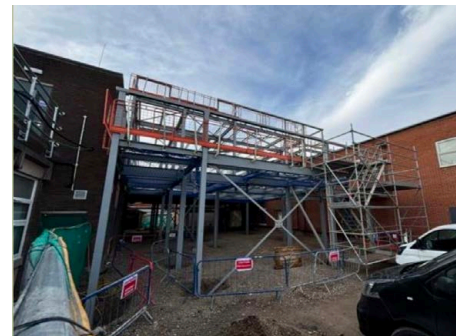
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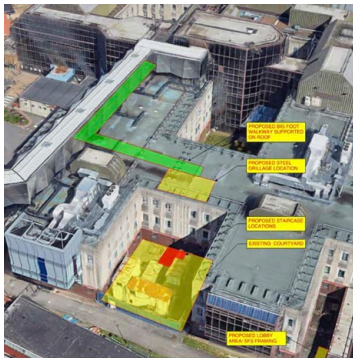
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10



3



7



11



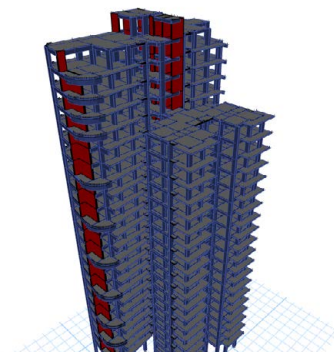
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12



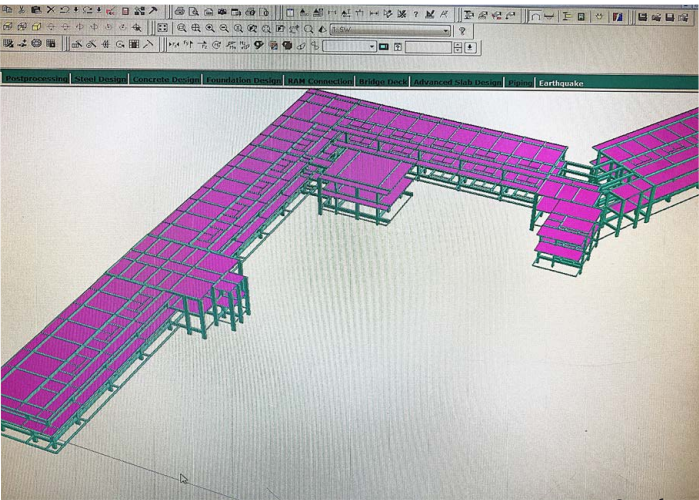
ECO RESORT

Location: Rishikesh, India
Year- July 2022
Status: Completed

Eco Resort is a 10,000 m² hotel in the Himalayan region of India. The site falls within Seismic Zone V, as classified by IS 1893-1. Due to seismic constraints, the height of the structure is limited to two stories. The resort is situated on sloped terrain, presenting significant engineering challenges—most notably the design of an 8-meter-tall cantilever retaining wall during the temporary condition and design of the structure in the sleep ground.

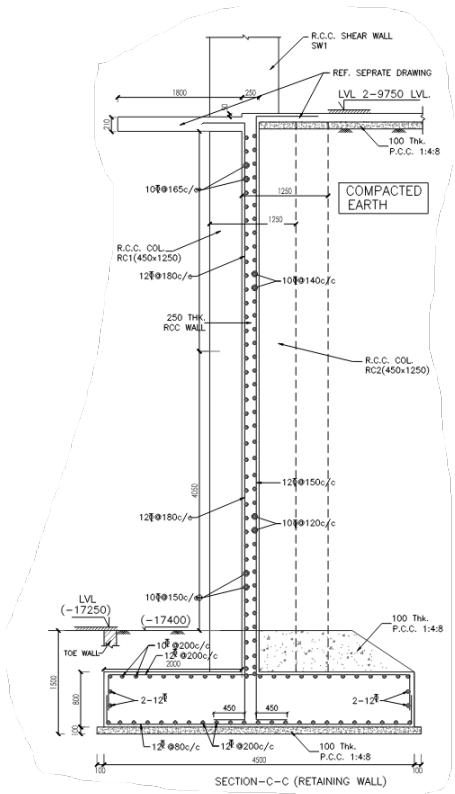
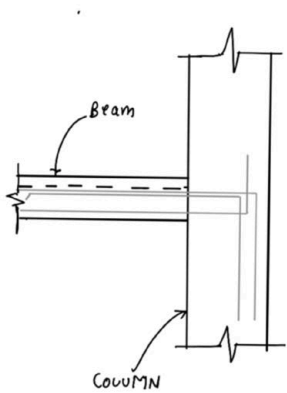
As part of the structural design team, I collaborated closely with the Senior Engineer to meet critical deadlines for Stage 5 deliverables, which included both structural drafting, framing and design work. Structural analysis was performed using STAAD.Pro, while detailed design calculations were completed using custom in-house Excel tools. All structural models were independently validated through hand calculations to ensure accuracy and code compliance.

This project provided valuable exposure to Stage 5 construction processes and deepened my understanding of seismic design principles, terrain-adaptive structural systems, reinforced concrete design and interdisciplinary coordination. Additionally, it significantly strengthened my teamwork, technical communication, and analytical skills.

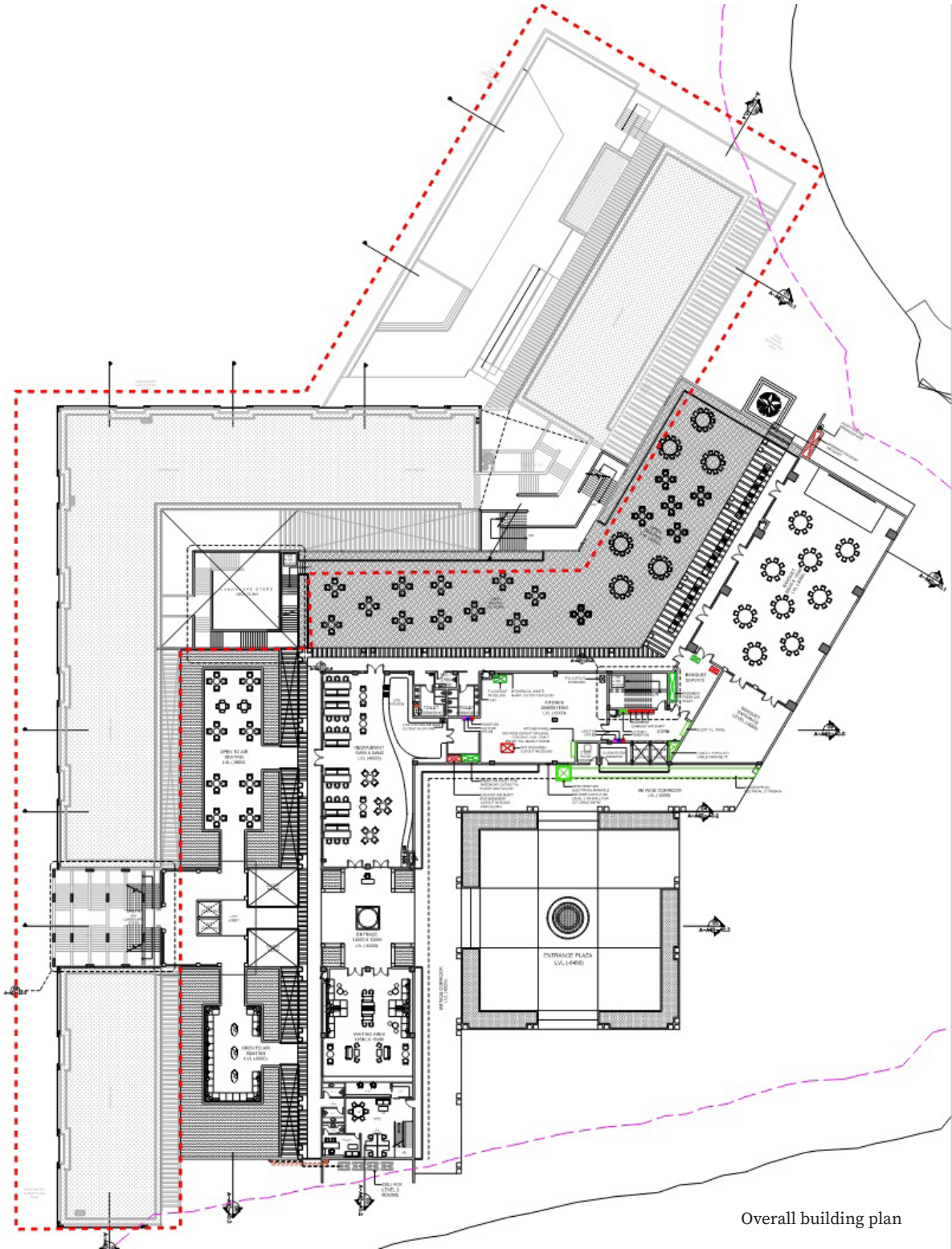
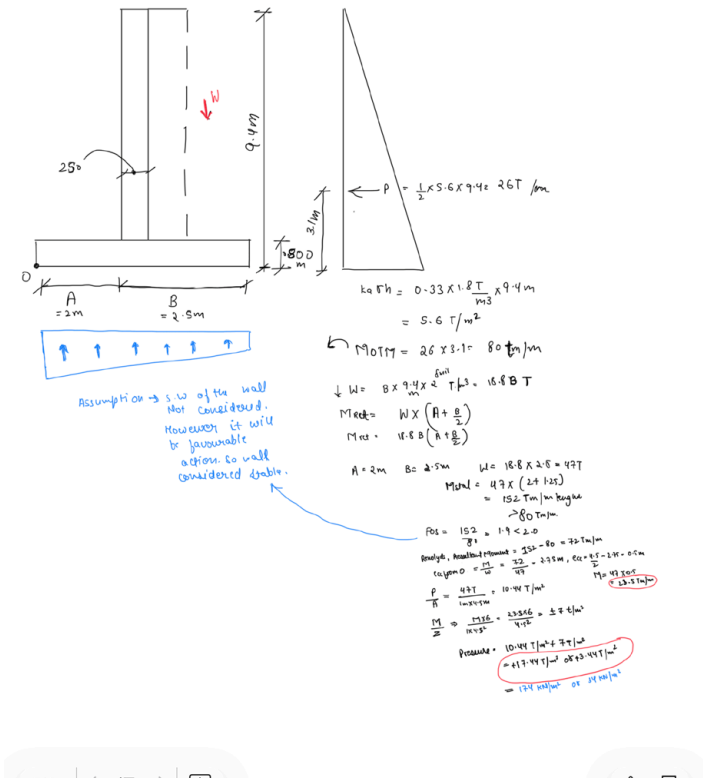


Staad pro main model

Beam to column- Moment connection

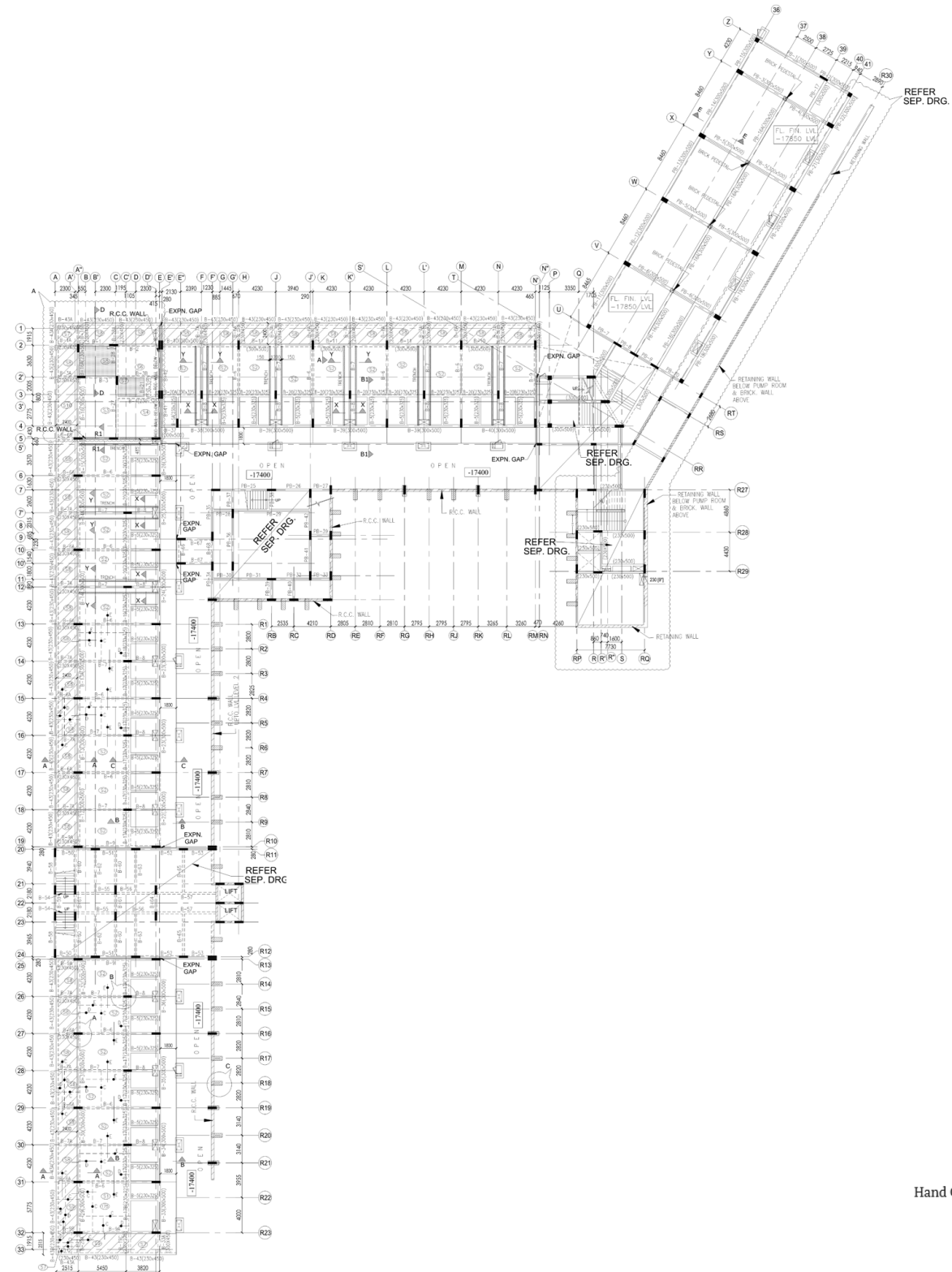


Counterfort cantilever retaining wall

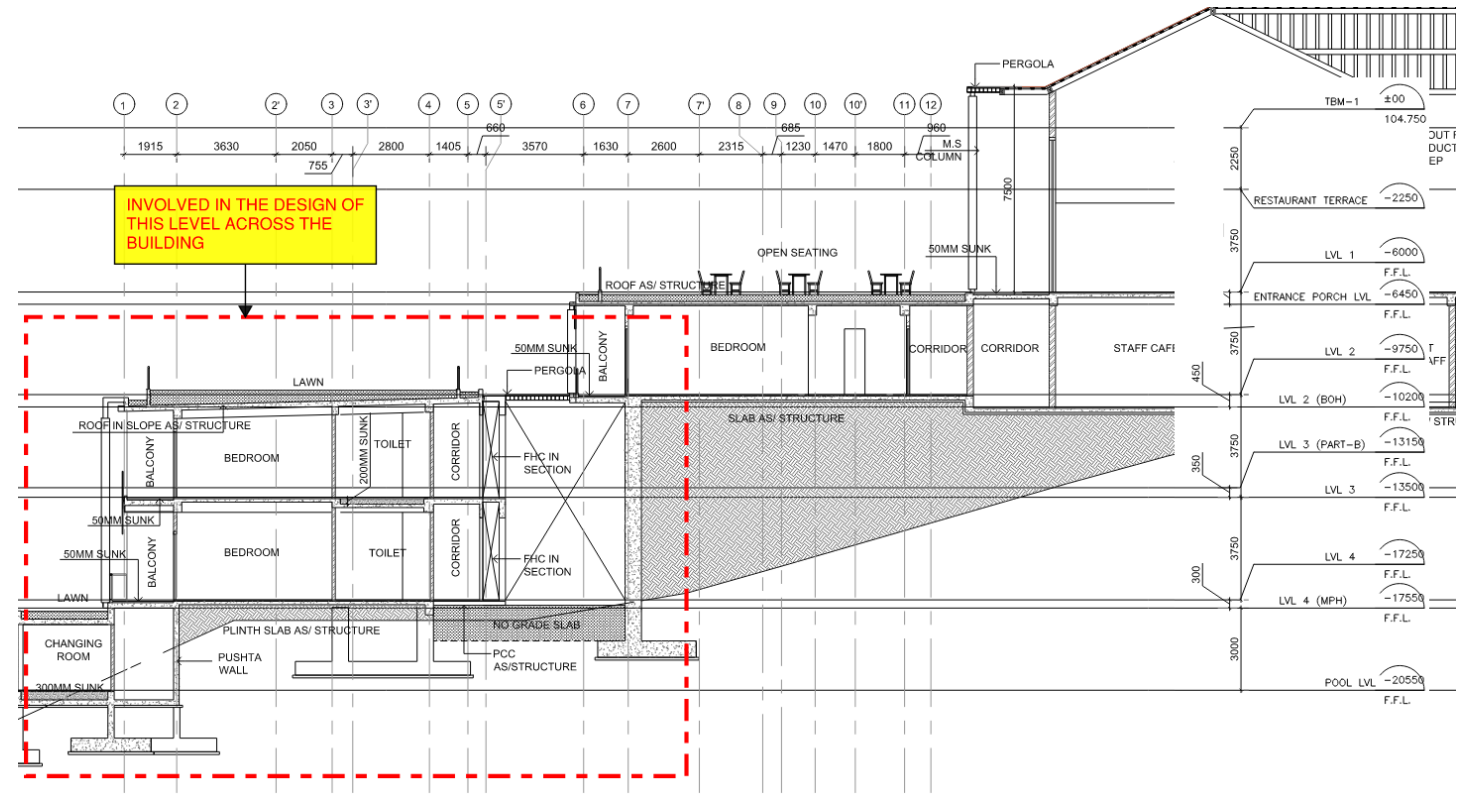


Overall building plan

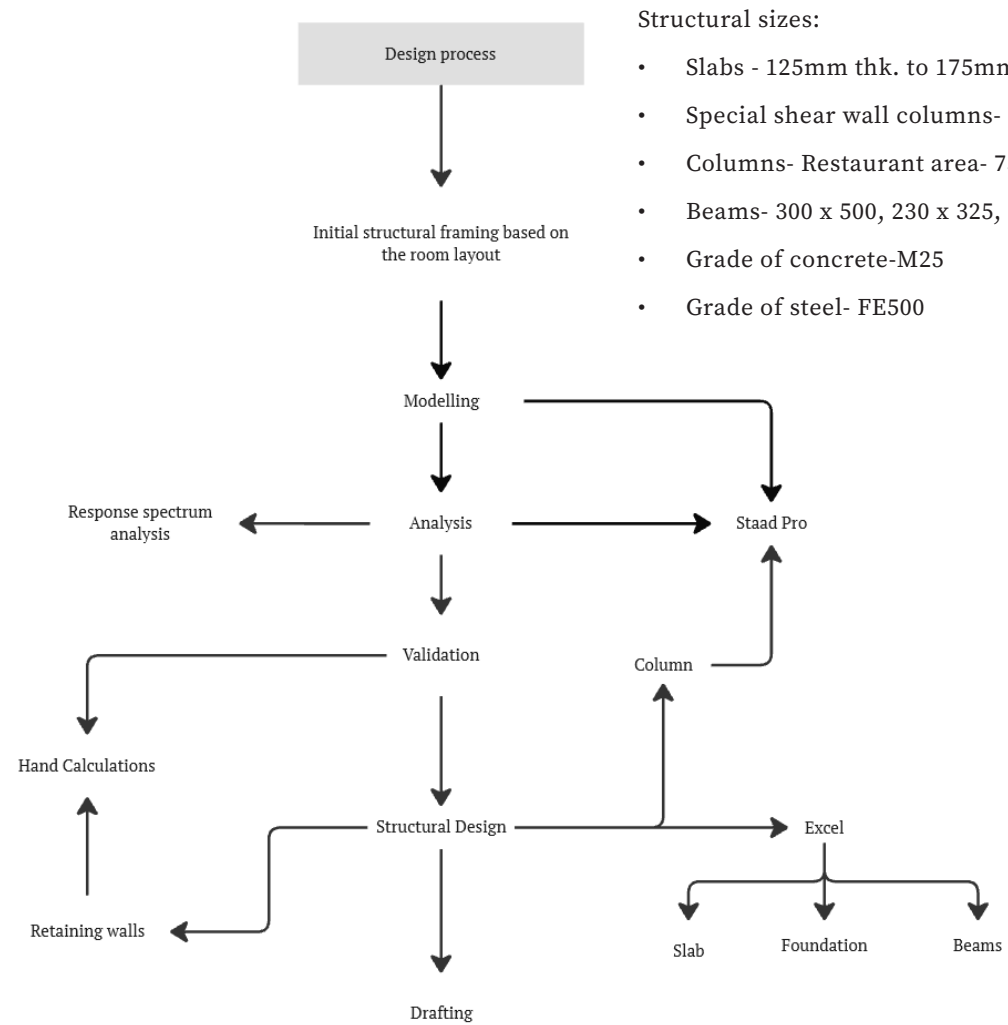
ECO RESORT



Plinth level building framing, Level -17550



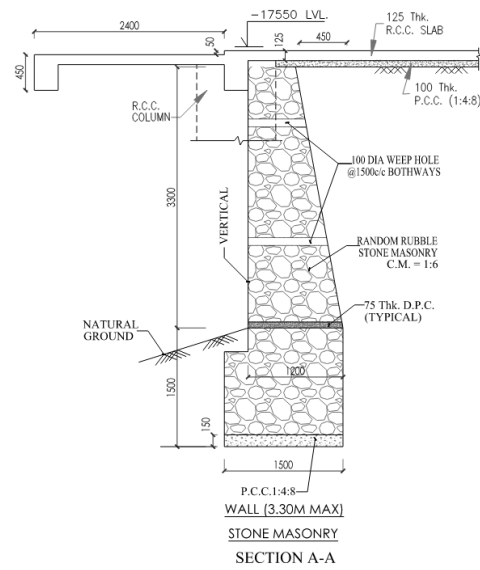
Building cross section



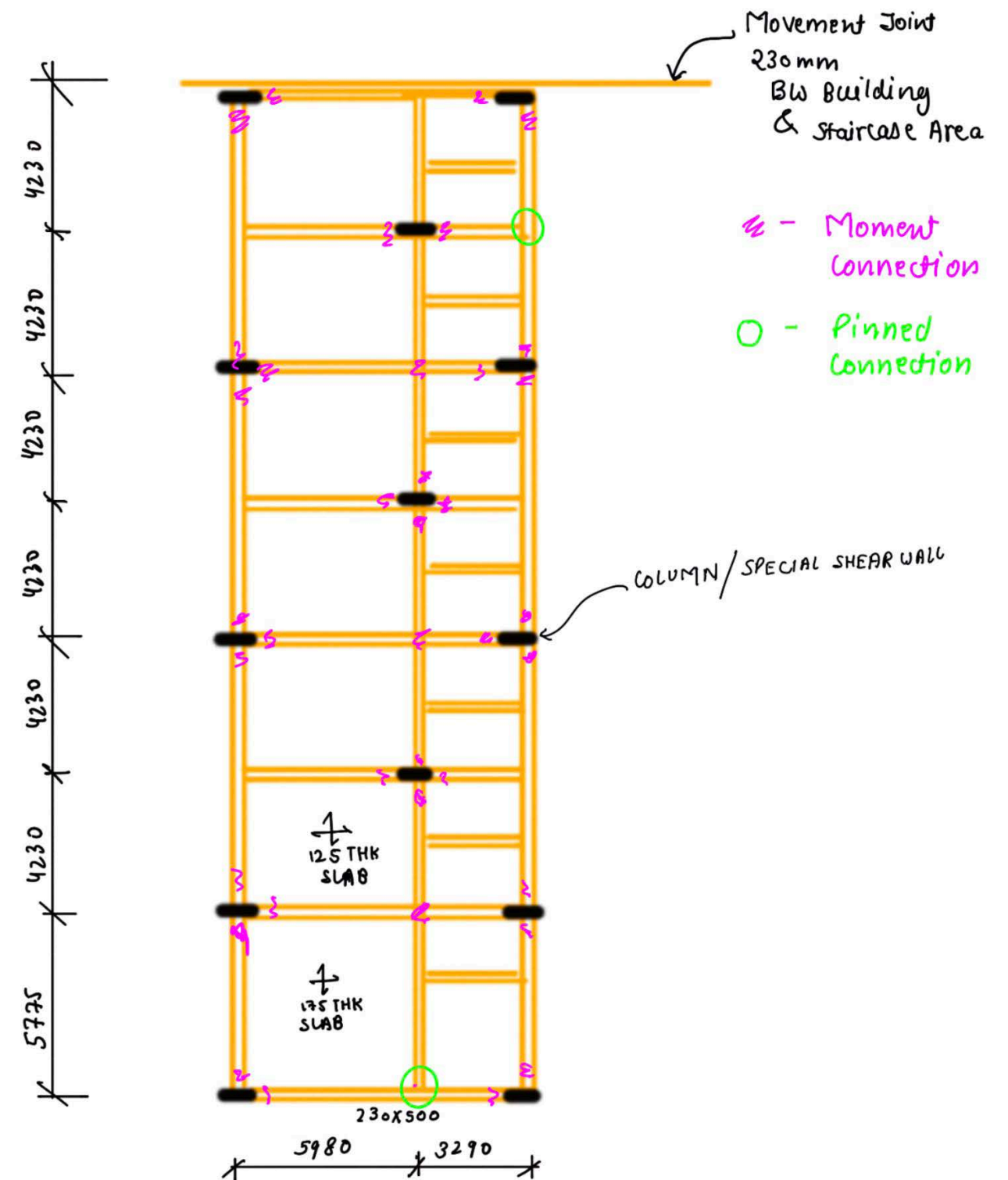
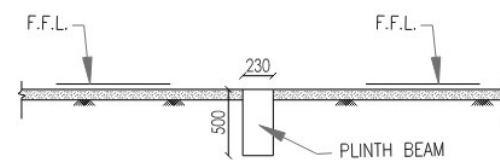
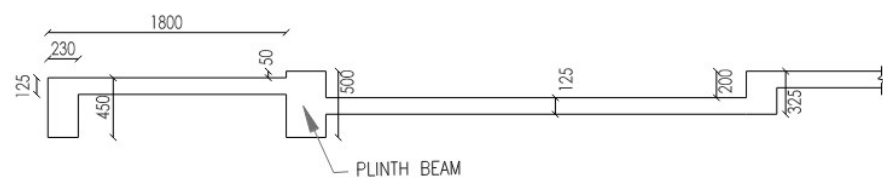
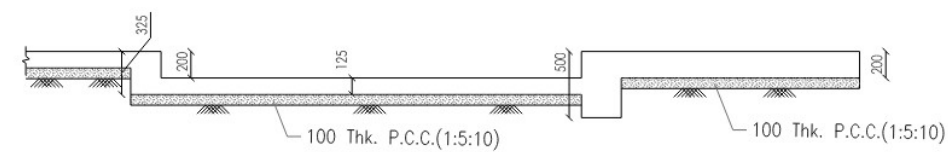
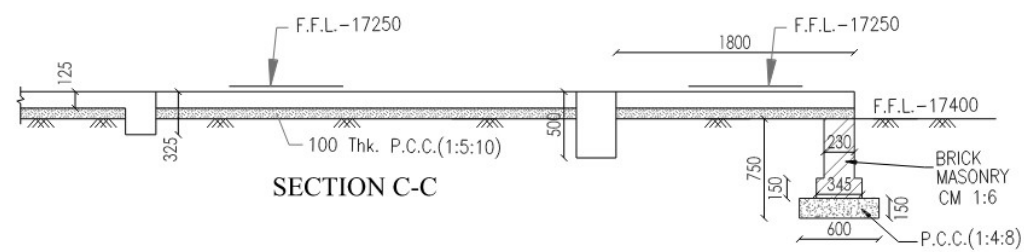
Structural sizes:

- Slabs - 125mm thk. to 175mm thk. 2 way slab between downstand beams
- Special shear wall columns- 1000 mm long x 230mm wide and 920 mm long and 230mm wide.
- Columns- Restaurant area- 750x 450, 450 x 450, 450 x 350
- Beams- 300 x 500, 230 x 325, 230 x 500
- Grade of concrete-M25
- Grade of steel- FE500

ECO RESORT



3m high stone masonry retaining wall with steel wire meshing. Stones for the masonry wall were sourced locally.



Framing arrangement indicating the fixities

Various section along the building slab and beams

MANCHESTER ROYAL INFIRMARY

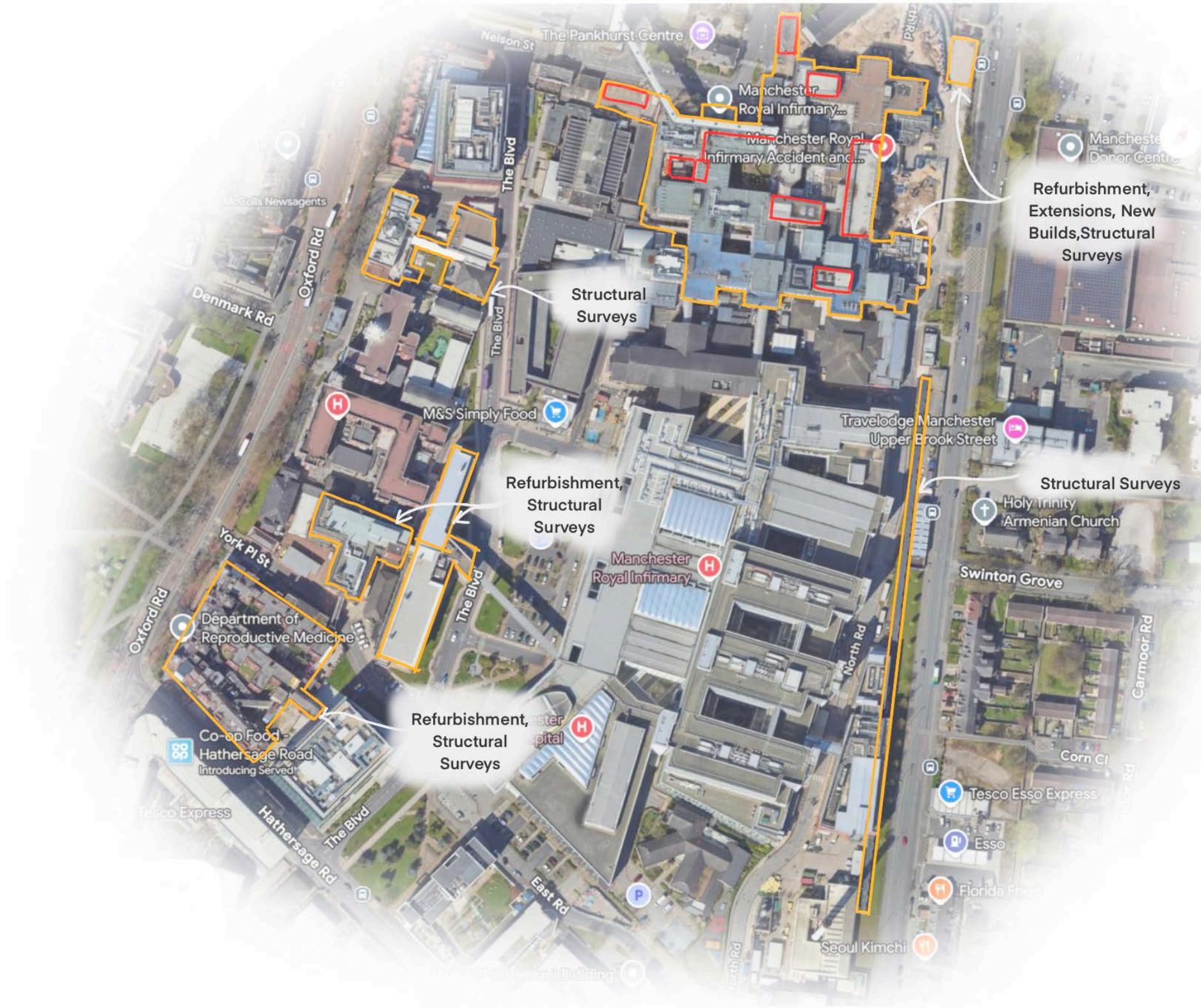
Location: Manchester
Year- 2023- Ongoing
Status: Under construction

Projects:

- Plantroom 2,3 and 5 Modifications
- Phase 2 RAAC surveys
- OSM Masonry inspections
- Refurbishment of the wards
- A&E Mechanical and Electrical supply routes (Permanent and Temporary)

My Role:

- Carrying out feasibility study and writing down the reports
- Structural surveys
- Concept design development
- Meeting with client and the contractor on site for discussion
- Modeling/ guiding junior engineer with modeling
- Checking drawings, reviewing calculation document
- Co-ordination with the various members of the design team



MRI PLANTROOM 2,3,&5 REFURBISHMENT

Location: Manchester
Year- 2025- Ongoing
Status: Stage 4

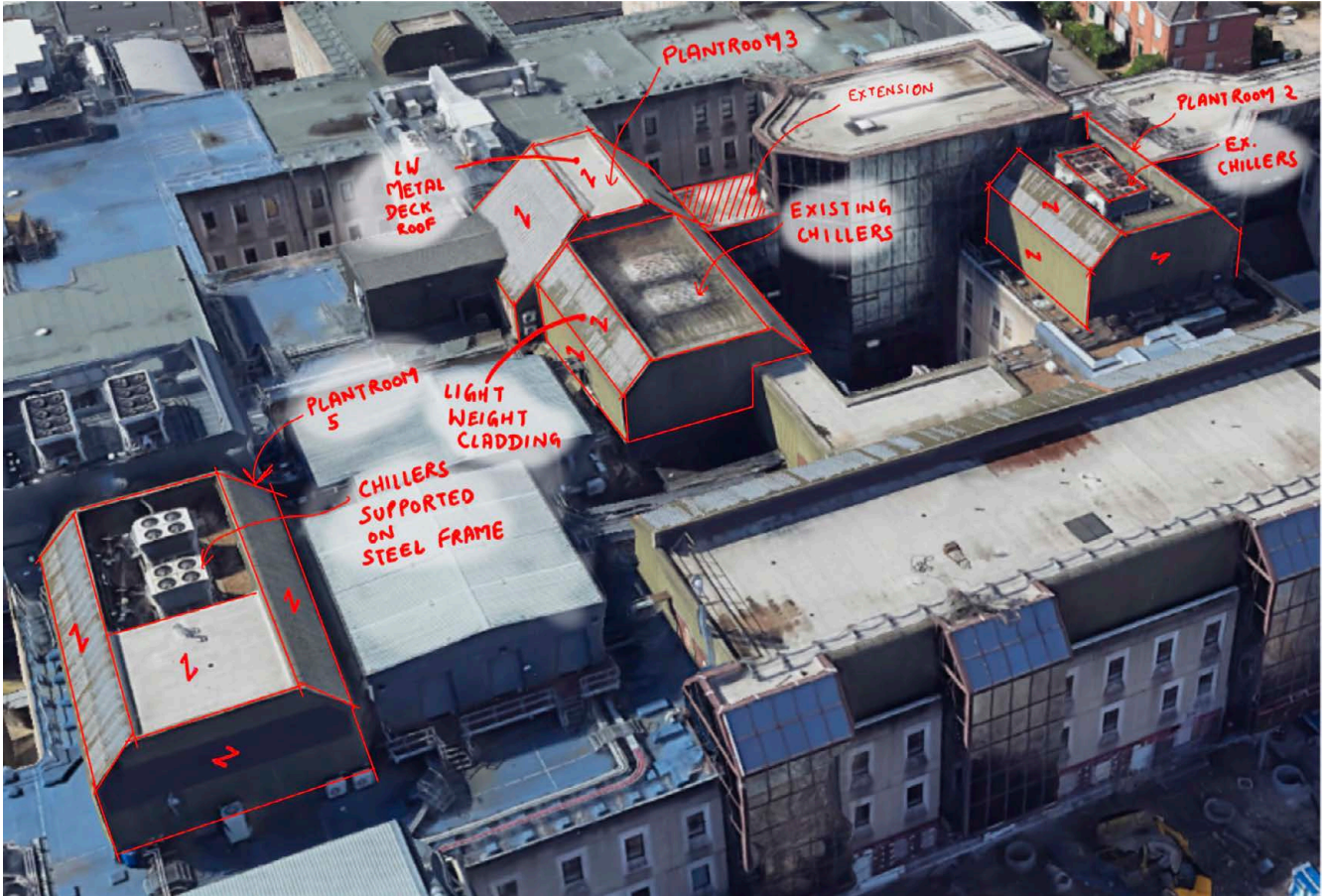
During the initial inspection, RAAC planks were identified in the roof and wall of Plantroom 2,3 and 5. The existing structure comprises a steel frame, moment resisting in the lateral direction and braced frame in longitudinal direction. Light weight metal deck cladding spans between cladding rails supported by the columns.

The overall scheme involved replacing the RAAC roof structure, upgrading the existing chillers and modifying the supporting frame to meet new operational requirements. The design was developed to minimise disruption to the live plantrooms during construction.

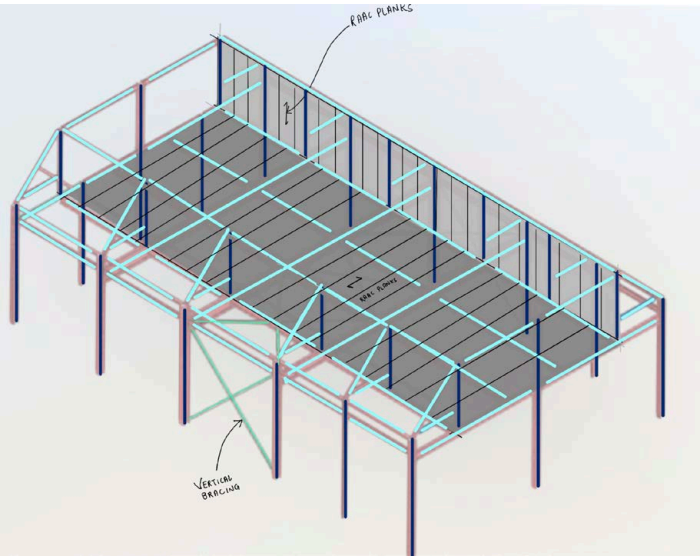
My Role (Stage 1-4)

- Conducted the detailed feasibility study and surveys of the existing structure.
- Assessed the existing steel frame and validated structural capacity by hand calculations and Tekla structural designer.
- Performed natural frequency check and sketch the modification intent by hand
- Surveyed adjacent structures to evaluate their suitability for supporting chillers during interim stages.
- Produced design intents for temporary works and contributed to logistics planning
- Attended design team meetings and site visits throughout the project.

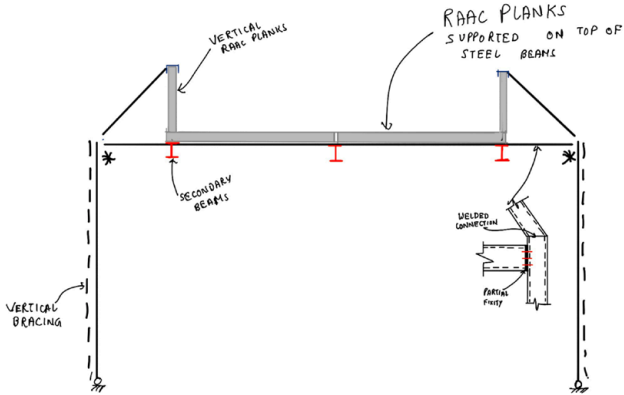
I learned about RIBA Stage 1 to Stage 4 design development, things to look during the survey of existing structure, simplified modeling of the structure with assumptions, logistics to site and design & build contract.



Overall site plan



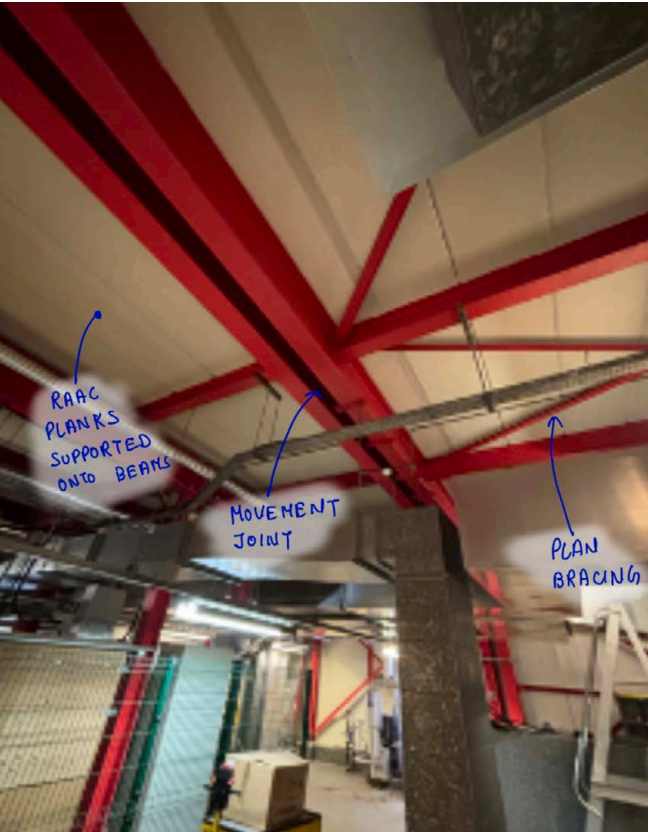
Plantroom 5- Existing structural arrangement



Existing RAAC Arrangement

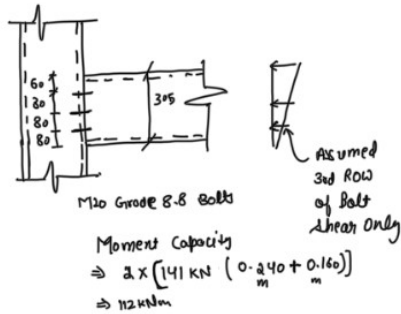
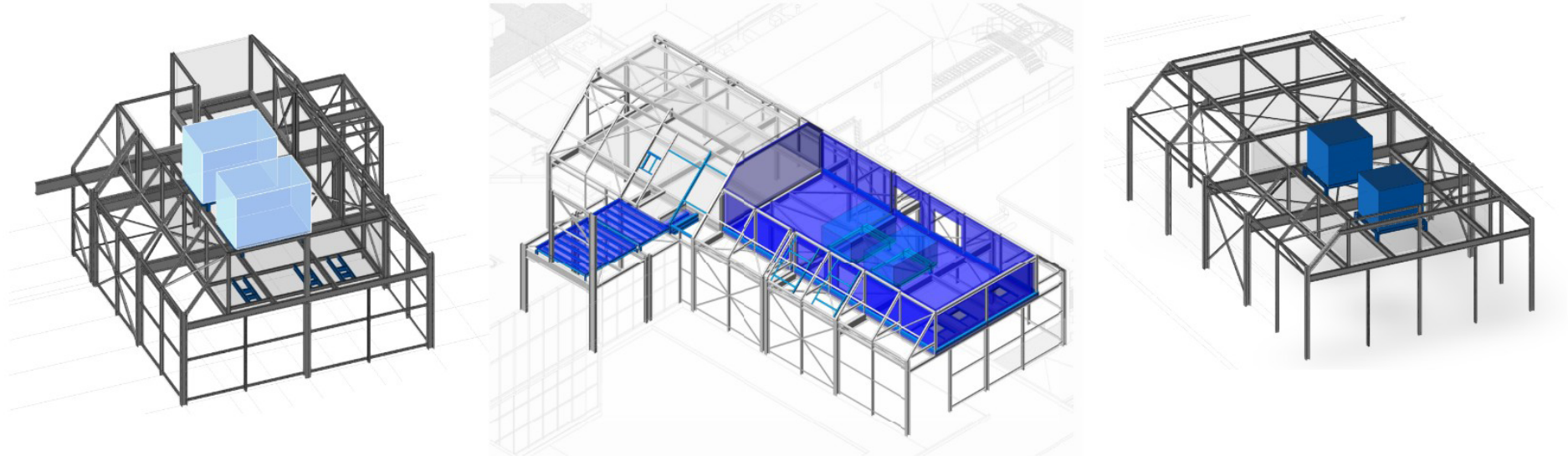
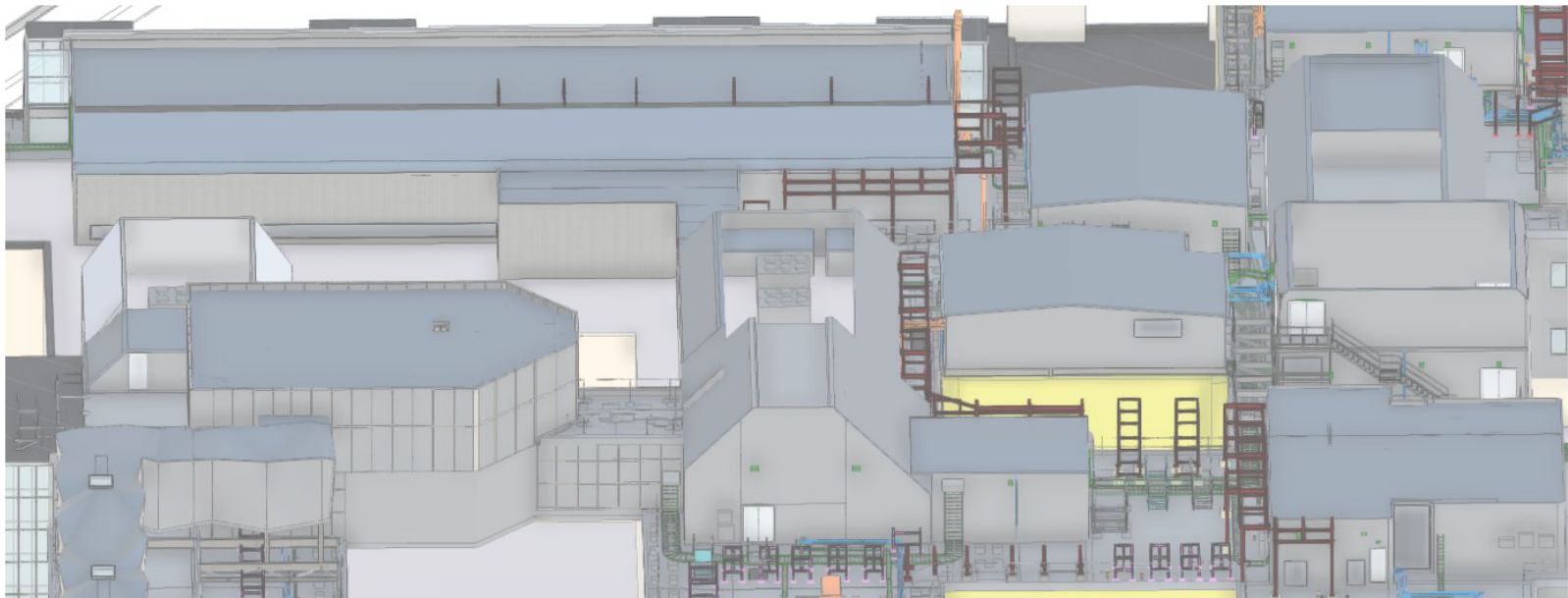


Plantroom 3- Moment connection



Typical moment joist in the structure

MRI PLANTROOM 2,3,&5 REFURBISHMENT



EXISTING BEAMS
CONSIDERED FIX

CONSERVATIVELY,
NATURAL FREQUENCY = $\frac{18}{\sqrt{6}}$

EXISTING LOAD = 2.67 kN/m^2
PROPOSED LOAD = 0.50 kN/m^2
 $\downarrow$
LIGHT WEIGHT METAL DECK ROOF

MRI EDUCATION NORTH

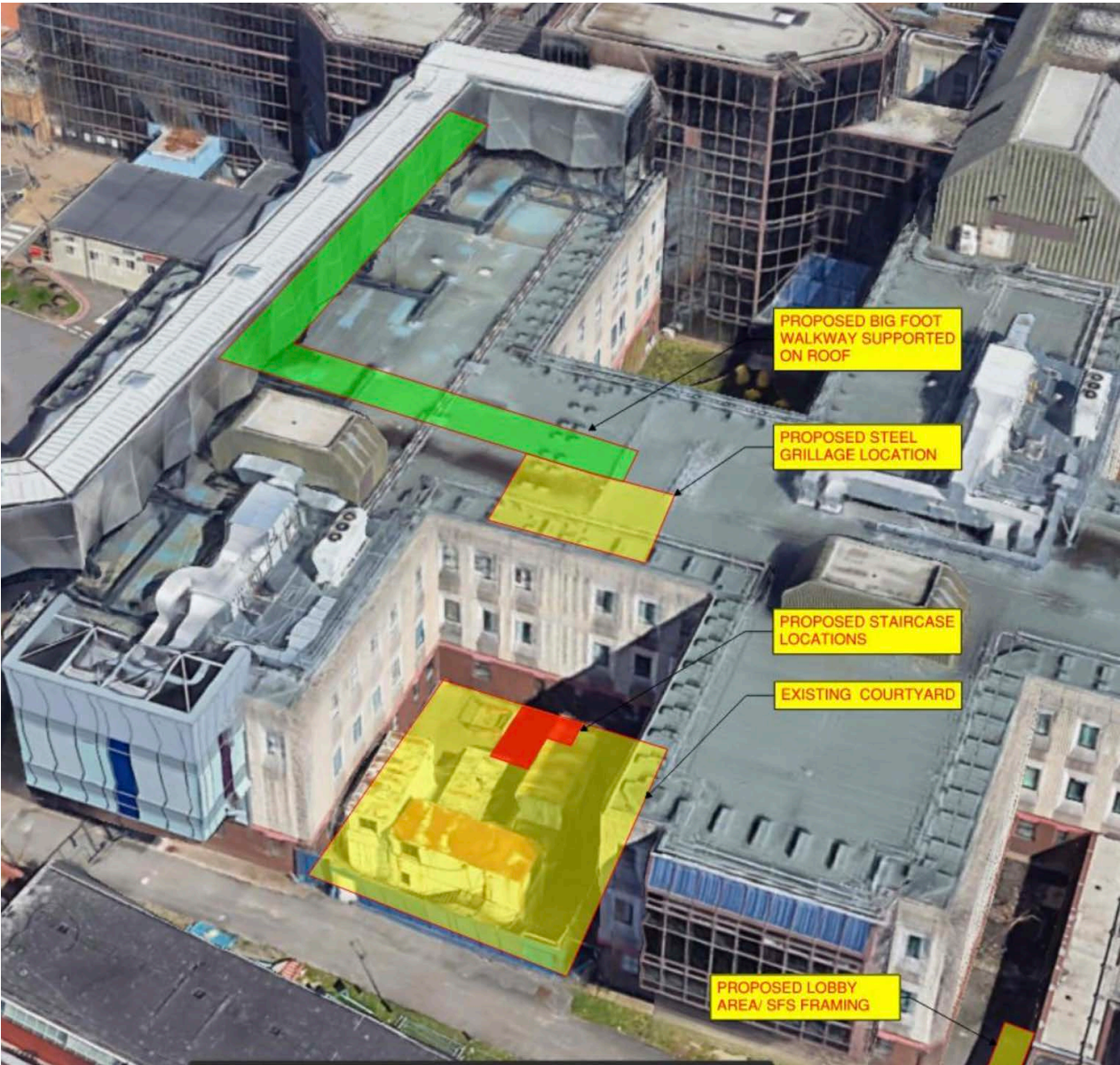
Location: Manchester
Year- 2025- Ongoing
Status: Stage 4

Additional access was required to reach Plantroom 2 for M&E replacements. The proposal included a new external staircase within the courtyard and a steel grillage platform over the existing roof structure. The grillage was designed to bear on the existing davit arm supports.

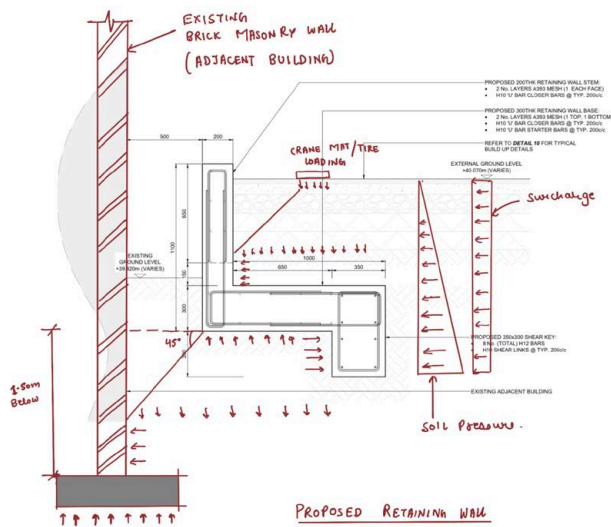
What I did as a part of the team:

- Conducted feasibility studies and structural surveys of the Phase 1 building.
- Assessed the existing concrete frame and validated capacity through Tekla modeling and hand calculations.
- Developed concept designs, temporary works proposals, and logistics strategies
- Assessed the existing brick masonry wall and basement, proposing a new retaining wall set 500 mm away to prevent excessive loading from crane outriggers.
- Prepared design intents for protective systems to safeguard the existing structure, including bollard protection.
- Participated in site visits and design team meetings throughout the project.
- Drafted initial proposals for ground investigation requirements such as soil contamination testing, trial pits etc.
- Completed the NBS specification document.

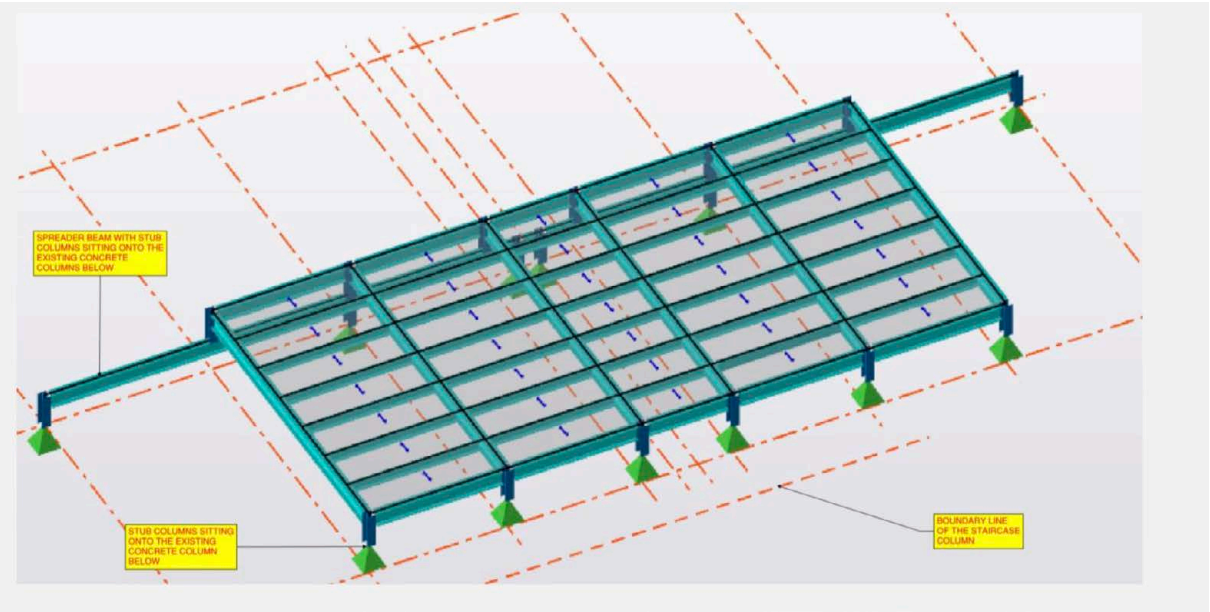
Major challenge on site to understand the existing below ground services and co-ordinate the existing structure around it. And to make inform the client regarding the reduced easement zones around the existing drains.



Overall Project plan

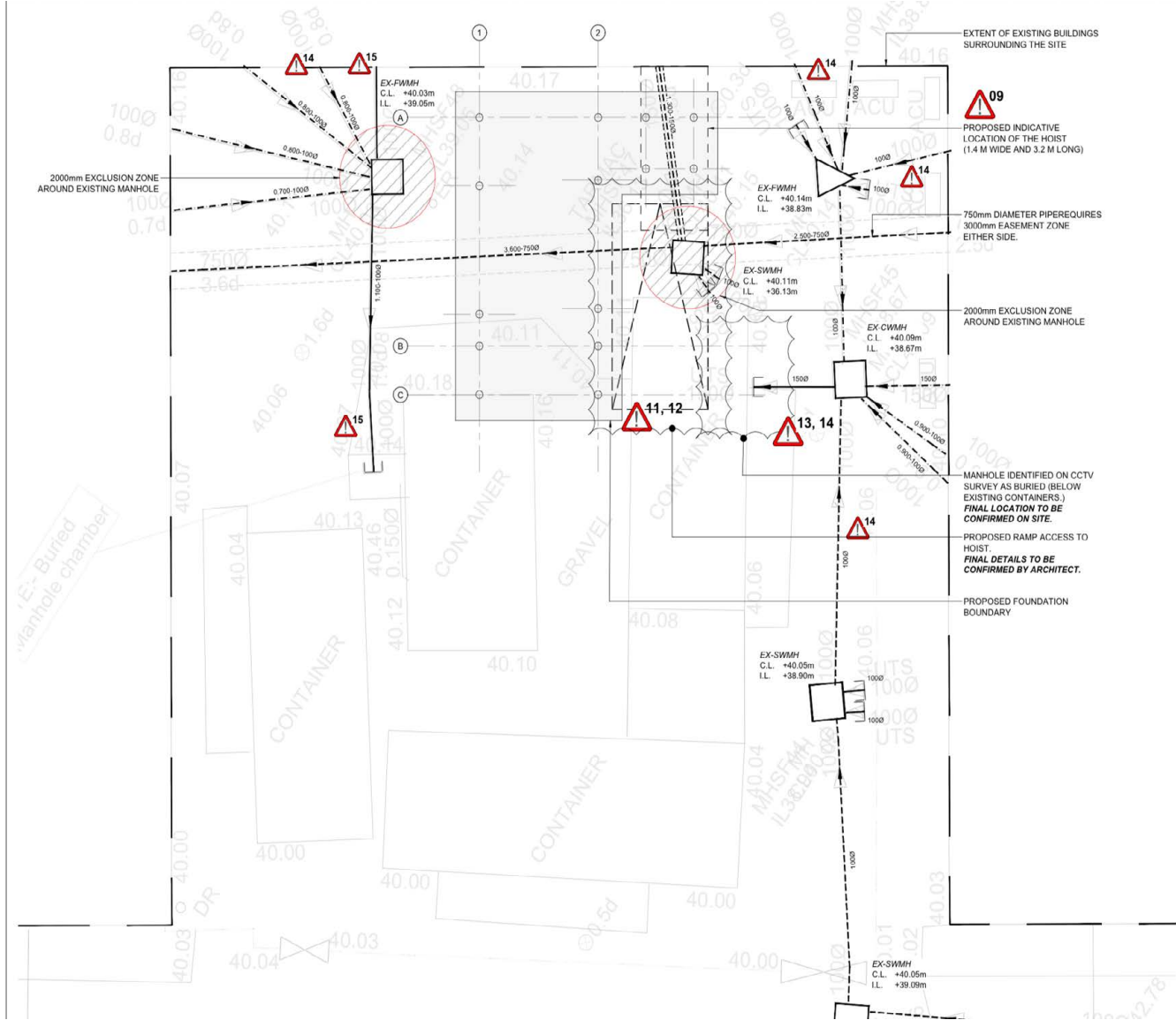


Retaining wall- Loading diagram



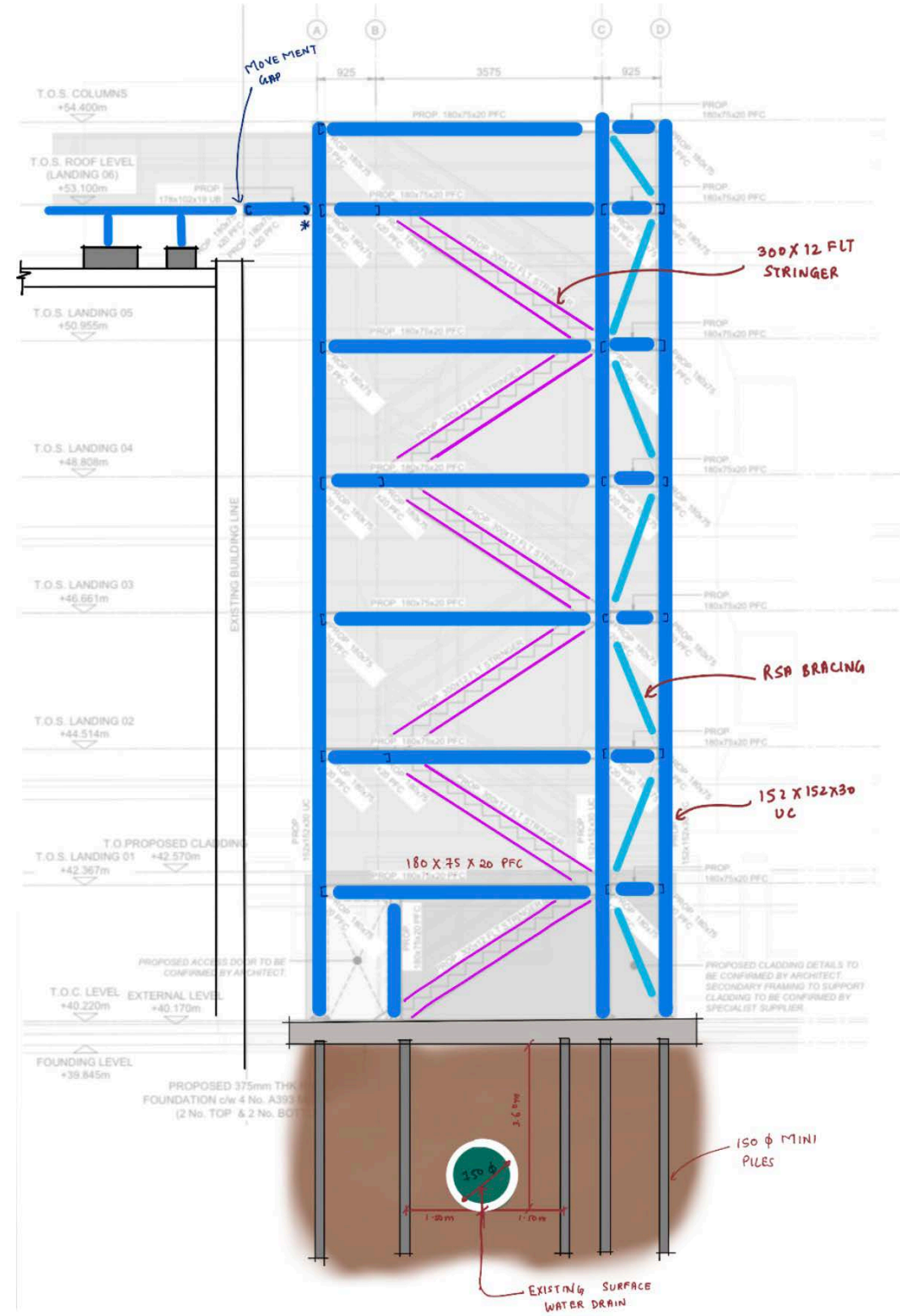
Steel Gantry Structure

MRI EDUCATION NORTH



Foundation layou

Proposed foundations were strategically co-ordinated around the existing 750 diameter drain pipe located at the depth of 3.6m below the existing ground level. 1.5m easement zone was provided on either side from the center line of the drain. The minipiles to go upto the depth of 7m.



Steel staircase section

RAAC PROJECTS (VARIOUS)

Location: NHS Hospitals, North West, England
Year- 2023- Ongoing
Status: Under construction

Reinforced Autoclaved Aerated Concrete (RAAC) is a lightweight material, widely used in the mid-1990s for constructing roofs in public buildings such as schools and hospitals. RAAC is approximately one-third the density of traditional concrete. Throughout the survey process, I participated in various phases of RAAC identification, remediation, and demolition. The work included:

- Phase 1: Visual identification of RAAC.
- Phase 2: Intrusive investigations, such as the Screwdriver Penetration Test.
- Phase 3: Detailed investigations, followed by emergency, short-term, and long-term remedial actions as per guidelines.

I worked closely with the Associate Structural Engineer on several tasks, including:

- Attending strategic meetings with the University of Manchester Foundation Trust and project managers to plan the survey while minimizing disruption to live hospital wards.
- Conducting desktop studies of buildings prior to on-site surveys.
- Carrying out phased site visits, inspecting one room or ward per week, following asbestos removal and ceiling strip-out, to assess plank conditions and recommend emergency propping at critical locations.
- Designing and drafting remedial solutions in steel and timber, including mid-span plank supports and bearing enhancements.
- Efficiently organizing survey notes and project files.
- Addressing contractor queries and assisting with the demolition strategy.
- Reviewing over-roof scaffolding drawings provided by the scaffolding contractor.
- Developing conceptual designs for a new timber roof with detailed interface considerations for the existing structure.
- Guiding a junior technician in drafting drawings while coordinating with the senior engineer.

Currently, I am leading the surveys and further schemes associated with RAAC such as roof replacement, refurbishment of the wards as a main engineer while acknowledging and communicating the decision/ solution with the senior engineer.



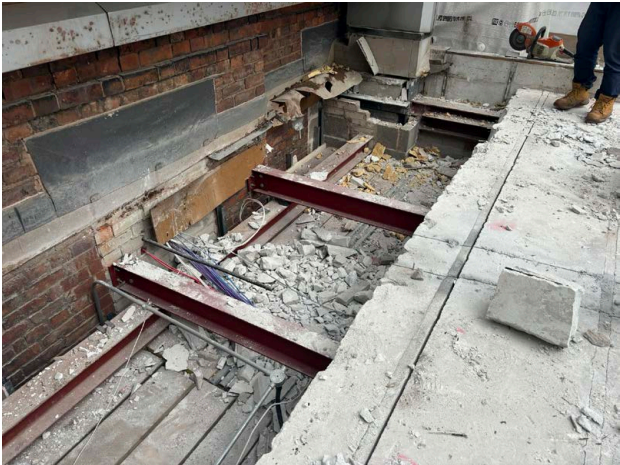
Historical Water Staining in the Planks



Emergency propping to support RAAC Planks



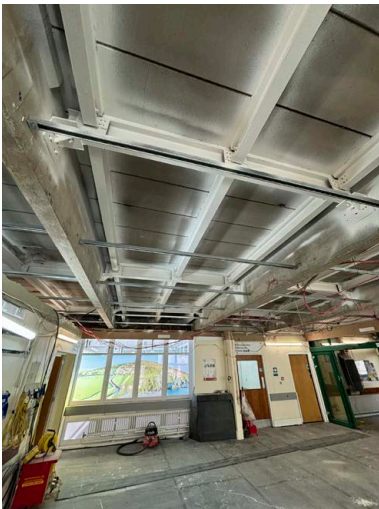
Exposed Reinforcement and Down-Pipes at RAAC Planks



Demolition of the RAAC Planks



Emergency Remedial
(Timber beam to support the roof)



Short- Term Remedial Installation (Steel beam to support the roof)

WYTHENSHAWE HOSPITAL

Location: Manchester
Year- 2023- Ongoing
Status: Under construction

Maternity Unit:

The project began with detailed inspections of RAAC planks to record defects and raise immediate concerns, followed by implementation of short-term remedial works. Inspections and remedials were completed across all areas, divided into four phases.

To facilitate the phased decanting of existing areas and enable remedial works, additional extensions and new temporary structures were required. These included a steel-framed corridor with timber floors and roofs supported on concrete strip and pad foundations, as well as raft foundations for modular clinical units. External works and below-ground drainage were developed to serve the new structures.

Following completion of external works, the scope extended to roof replacement for Phase 3. This involved coordination with the wider design team (Architects and M&E), internal demolition, temporary works to enable demolition, and ongoing site engagement. Responsibilities included weekly design coordination meetings, drainage proposals, regular site visits to review demolition and drainage works, and reviewing subcontractor drawings

External works also included new road enclosures and widening of a private roadway to a two-way route.

Cystics Roof:

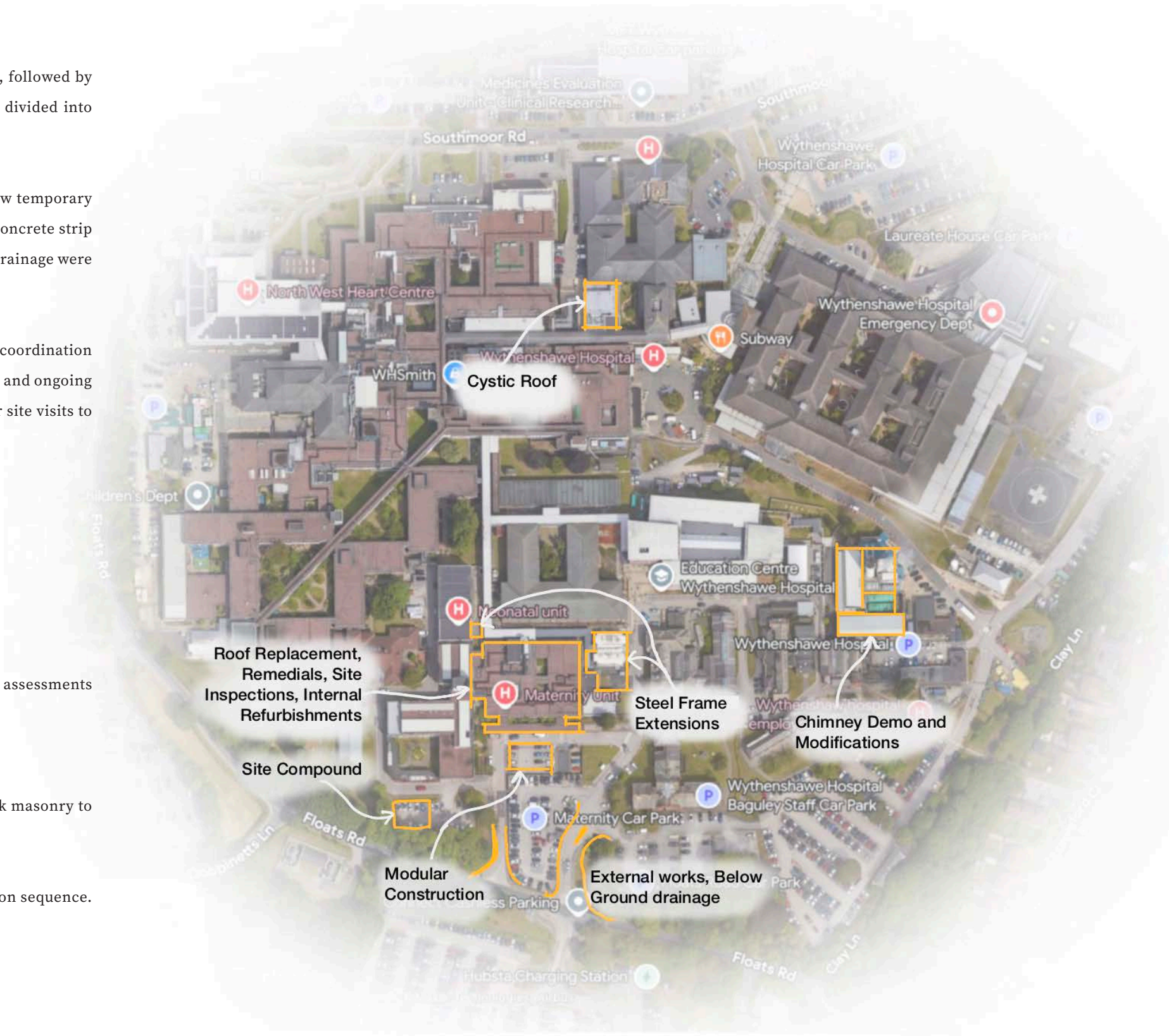
Installation of a new steel canopy over the existing structure

- Conducted site visits to verify existing archive information.
- Attended design team meetings and liaised with the steelwork fabricator.
- Validated the existing roof structure and reviewed subcontractor connection calculations, including load assessments from crane outriggers.

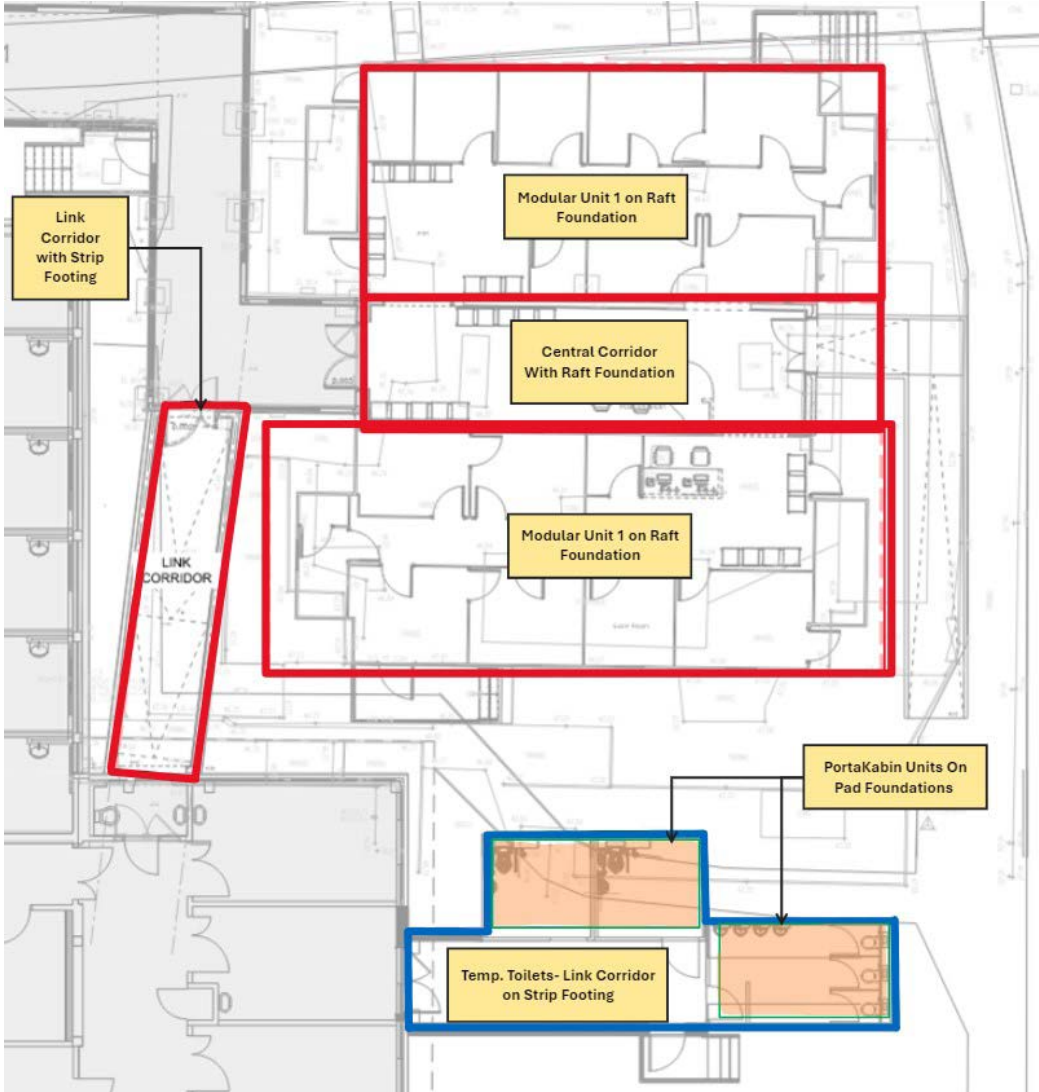
Chimney demolition:

Demolition of an existing chimney structure to a height of 15 m, following extensive removal of internal brick masonry to allow safe access.

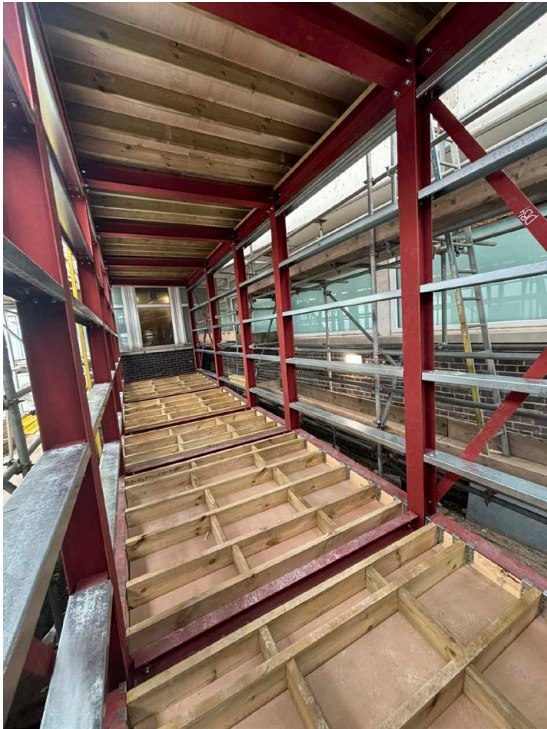
- Reviewed contractor demolition methodology and feasibility reports.
- Provided design intent for temporary works and new structural openings required as part of the demolition sequence.



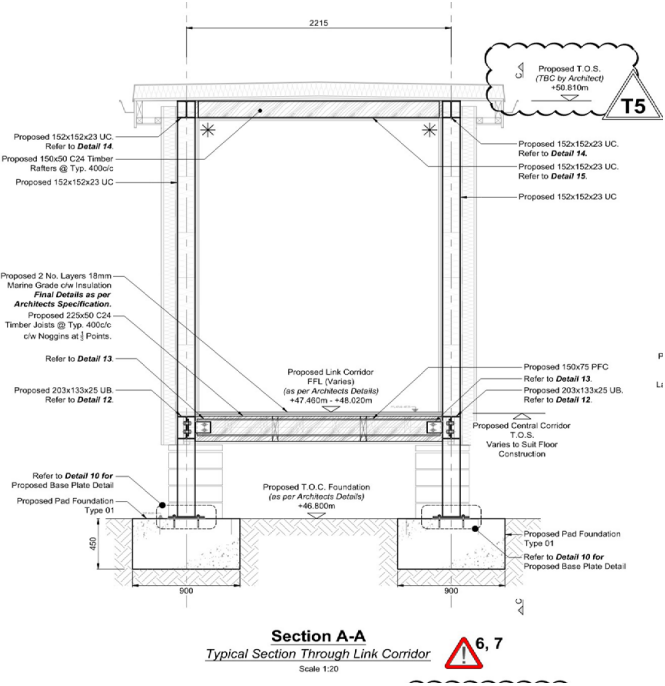
WYTHENSHAW HOSPITAL



Overall Site Plan (Extension)



Photos of the Link Corridor taken during construction



Demolition of roof structure



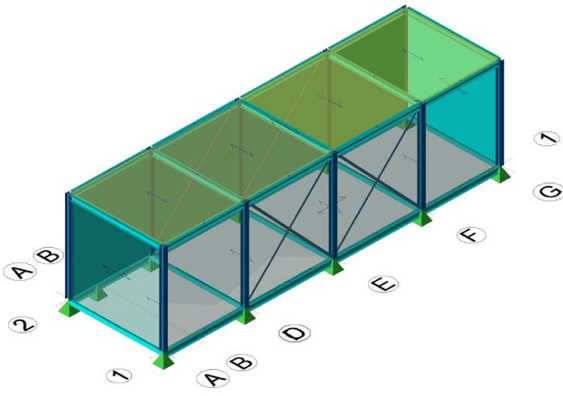
New timber roof construction (200 x 50 C24 Timber joist at 400 c/c)



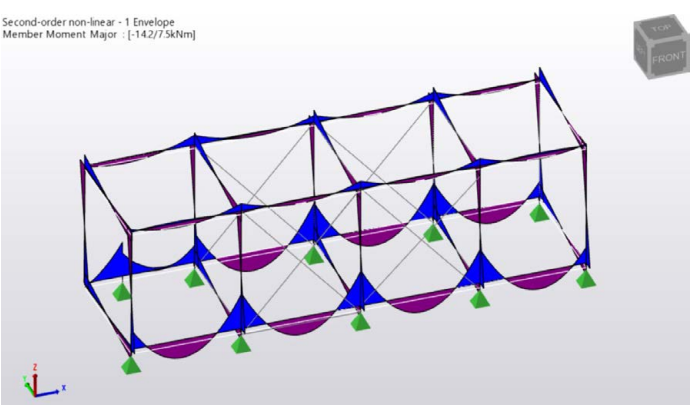
Temporary units sitting on the new foundation



Temporary clinics with Ramp and steel frame corridor



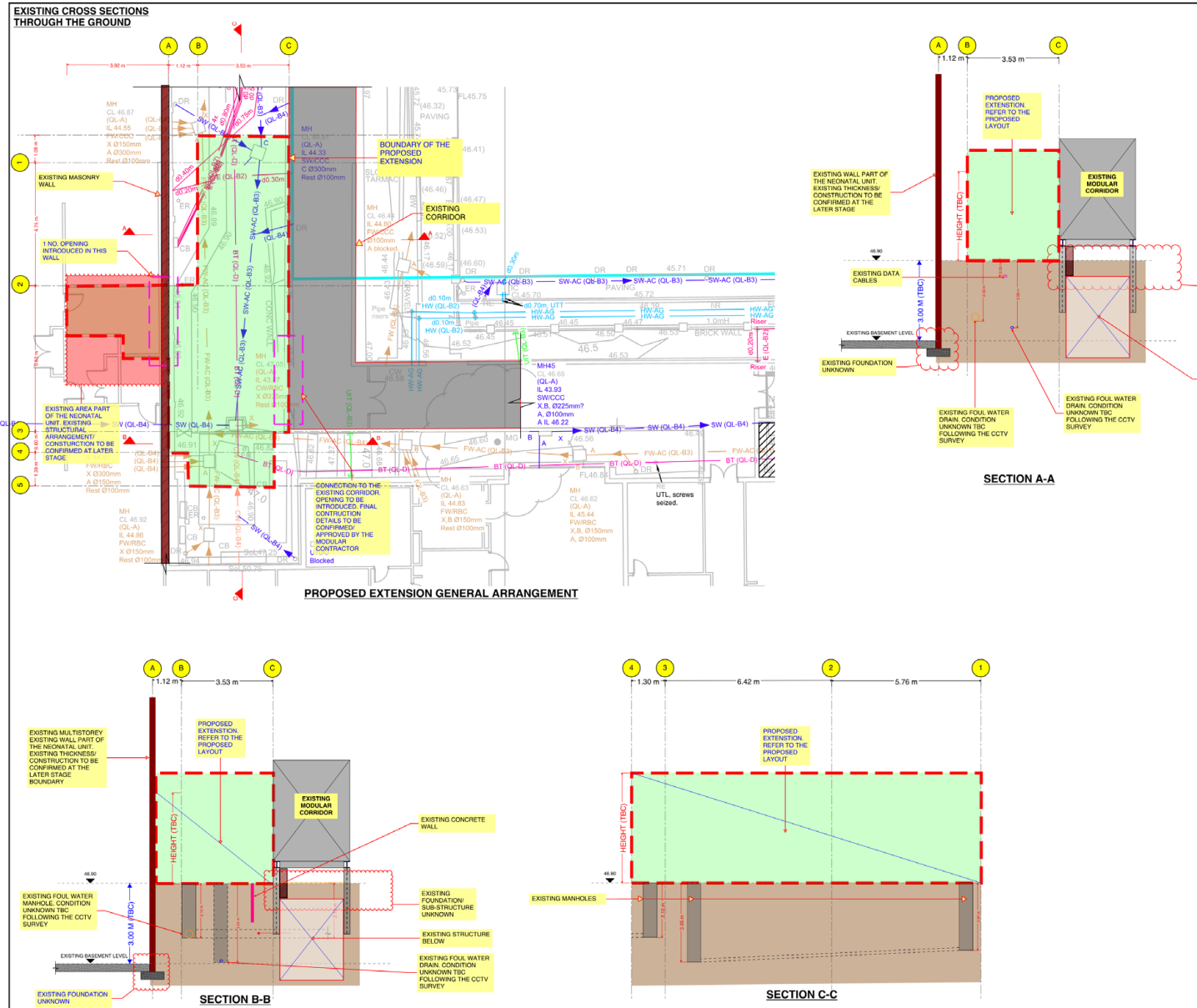
Corridor Tekla Model



Bending model diagram (TSD Output)

WYTHENSHAW HOSPITAL

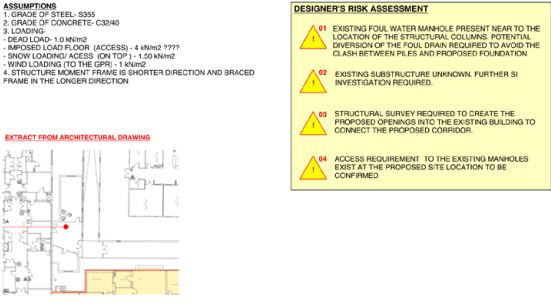
CONCEPT DESIGN



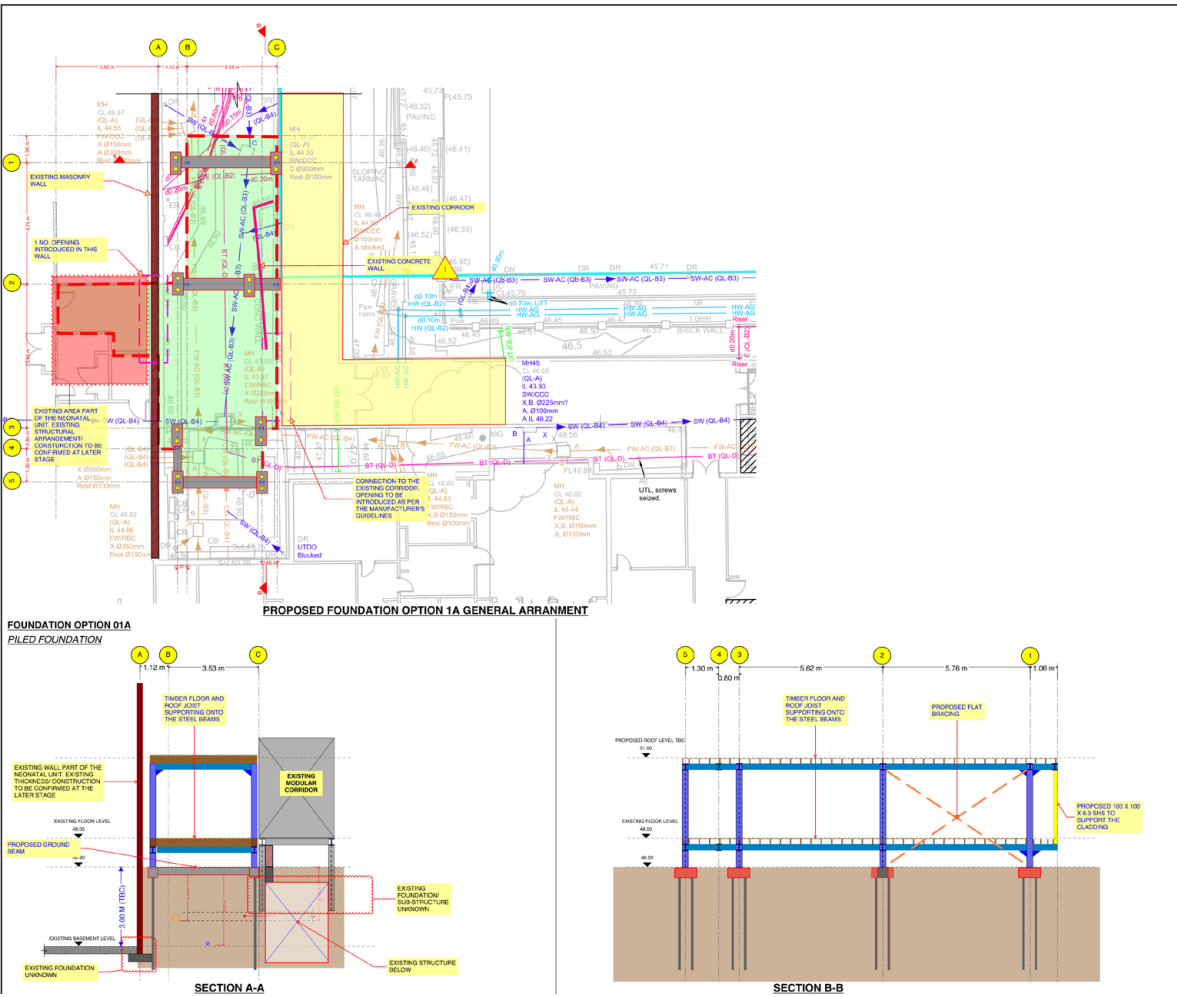
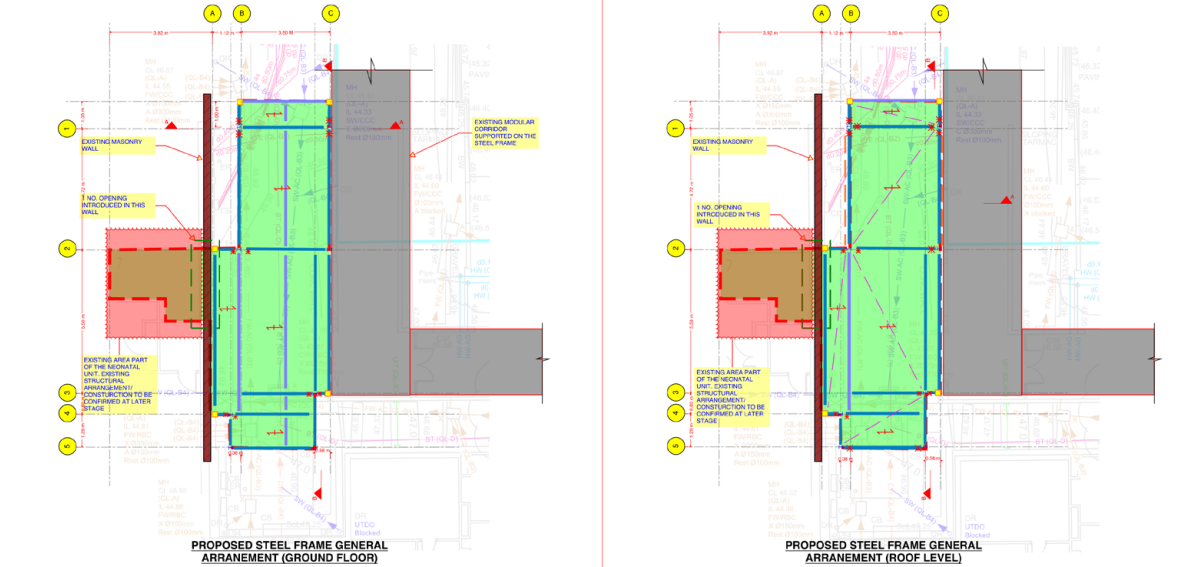
Location of existing services with cross section in both direction

New steel frame extension to connect 2 different corridor located in the separate buildings. New extension is located in the restricted location and surrounded with 3No. building with significant amount of the below ground services. Feasibility study was carried out to scope the requirements of site investigation. In addition, 2 no. conceptual framing opening were created including 2 different foundation layout as a initial proposal to the client for the discussion. Mudstone is located at the shallow depth.

OPTION 1- INDEPENDENT STEEL FRAME CONNECTING TO THE ADJACENT STRUCTURES WITH 2 NO. FOUNDATION OPTIONS



- SURVEYS REQUIREMENTS**
- TOPOGRAPHICAL SURVEY (RECEIVED)**
 - GPS SURVEY (RECEIVED)**
 - BELOW GROUND DRAINAGE CCTV SURVEY TO INCLUDE:**
 - DRAINAGE LAYOUT WITH SURFACE WATER AND FOUL IDENTIFICATION, INCLUDING PIPE SIZES, GULLEYS, ACO DRAINS ETC
 - LOCATION OF THE MANHOLES / PUMPING CHAMBERS AND THEIR DIAMETERS, INVERT AND COVER LEVELS, FLOW DIRECTION ETC
 - CONDITION / MANHOLE SPEC SHEET
 - CCTV SURVEY INDICATING CONDITION OF DRAINS - WNCAM SURVEY WITH CLEARLY MARKED GULLEYS, ACO, MHS / RUNS / DEFECT CHARGES
 - DRAWINGS TO BE PROVIDED IN PDF AND DWG FORMAT (IF POSSIBLE), INCLUDING ALL THE ABOVE INFORMATION IN COLLATED REPORT
 - SURVEY EXTENT NEEDS TO GO TO THE MAIN SEWER LINE
 - MANHOLE CARDS
 - GEOTECHNICAL SITE INVESTIGATION**
 - FOR PILE FOUNDATION SOLUTION THEN 2 NO. WINDOW SAMPLING/ BORE HOLE UPTO 10M DEPTH TO DETERMINE SOIL PROFILE AND ALLOWABLE GROUND BEARING PRESSURE FOR MINI PILES
 - FOR RAFT FOUNDATION SOLUTION, 4 NO. TRIAL PIT ADJACENT TO THE EXISTING BUILDING (LOCAL TO PRIMARY COLUMN) TO DETERMINE THE DEPTH OF THE EXISTING FOUNDATION (HARD DUG)
 - FOR RAFT FOUNDATION SOLUTION TAKEN FORWARD, SHEAR VANE TEST REQUIRED TO DETERMINE THE SOIL BEARING CAPACITY
 - MATERIAL SAMPLING / TESTING TO INCLUDE - ASBESTOS, WASTE ACCEPTANCE CRITERIA, CONTAMINATION SUITE
 - SOIL GAS SURVEY REQUIRED TO IDENTIFY THE POTENTIAL GAS RELEASE FROM THE EXISTING SOIL

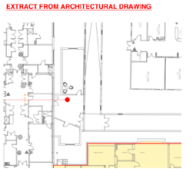


WYTHENSHAW HOSPITAL

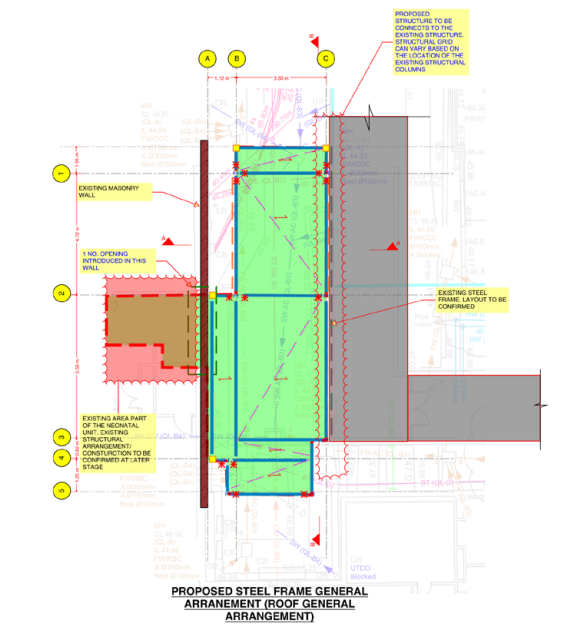
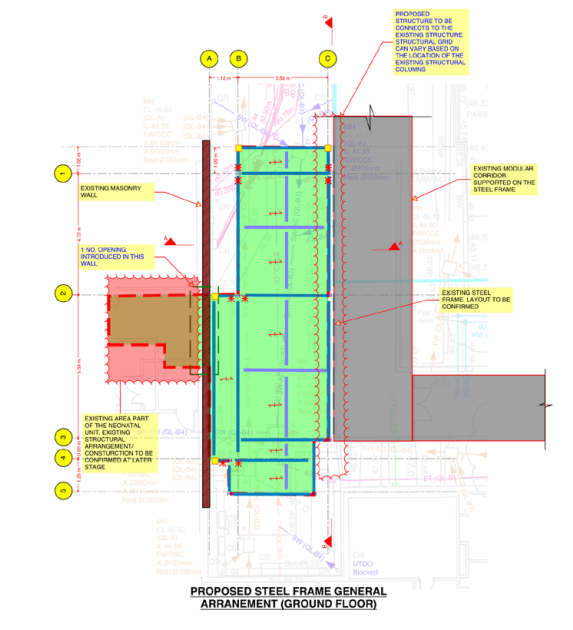
OPTION 2- STEEL FRAME SUPPORTED WITH THE ADJACENT STEEL FRAME CONNECTING TO THE ADJACENT STRUCTURES

ASSUMPTIONS
1. GRADE OF STEEL: S235
2. GRADE OF CONCRETE: C30/37
3. LOADING:
- DEAD LOAD: 1.0 kN/m²
- IMPROVED LOAD FLOOR (ACCESS): 4 kN/m²
- SNOW LOADING: ACCESS (ON TOP): 1.50 kN/m²
- WIND LOADING TO THE UPPE: 1 kN/m²
4. STRUCTURE MOMENT FRAME IS SHORTER DIRECTION AND BRACED FRAME IN THE LONGER DIRECTION

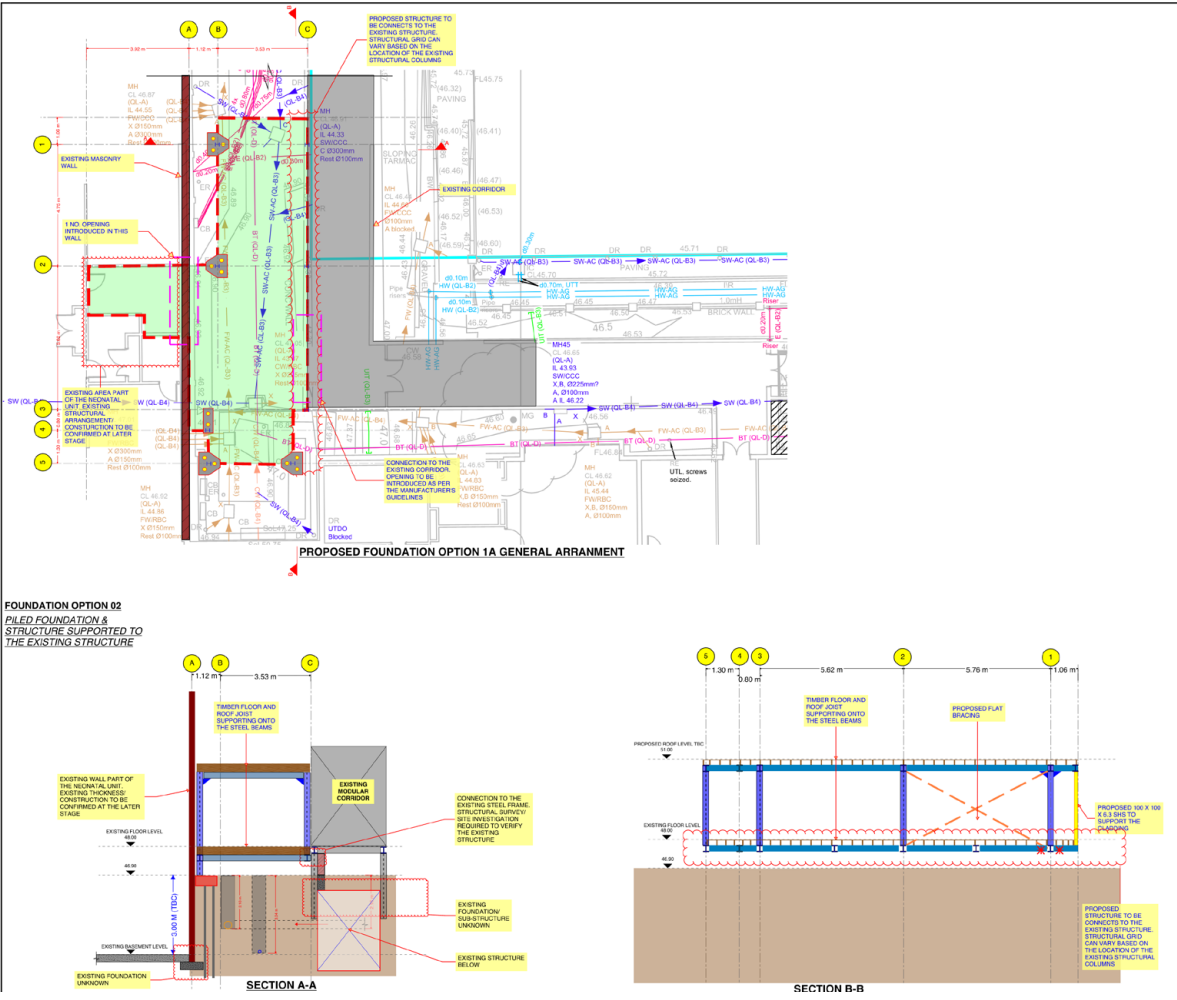
- DESIGNER'S RISK ASSESSMENT
- 01 EXISTING MANHOLES PRESENT ON SITE. CAREFUL CONSIDERATION SHOULD BE CARRIED WHILE CARRYING OUT THE EXCAVATION WORK TO AVOID ANY DAMAGE TO THE EXISTING MANHOLES.
 - 02 EXISTING SUBSTRUCTURE UNKNOWN. FURTHER SI INVESTIGATION REQUIRED.
 - 03 STRUCTURAL SURVEY REQUIRED TO CREATE THE PROPOSED OPENINGS INTO THE EXISTING BUILDING TO CONNECT THE PROPOSED CORRIDOR
 - 04 ACCESS REQUIREMENT TO THE EXISTING MANHOLES EXIST AT THE PROPOSED SITE LOCATION TO BE CONFIRMED



- SURVEYS REQUIREMENTS
- TOPOGRAPHICAL SURVEY (RECEIVED)
- GPR SURVEY (RECEIVED)
- BELOW GROUND DRAINAGE CCTV SURVEY TO INCLUDE:
- DRAINAGE LAYOUT WITH SURFACE WATER AND FOUL IDENTIFICATION, INCLUDING PIPE SIZES, GULLEYS, ADD DRAINS ETC
 - LOCATION OF THE MANHOLES / PUMPING CHAMBERS AND THEIR DIAMETERS, INVERT AND COVER LEVELS, FLOW DIRECTION ETC
 - LOCATION / MANHOLE SPEC SHEET
 - CCTV SURVEY INDICATING CONDITION OF DRAINS – WINCAM SURVEY WITH CLEARLY MARKED GULLEYS, ACC. MHS / RUNS / DEFECT CHANGES
 - DRAWINGS TO BE PROVIDED IN PDF AND DWG FORMAT (IF POSSIBLE), INCLUDING ALL THE ABOVE INFORMATION IN COLLATED REPORT
 - SURVEY EXTENT NEEDS TO GO TO THE MAIN SEWER LINE
 - MANHOLE CARDS
- GEOTECHNICAL SITE INVESTIGATION
- FOR PILE FOUNDATION SOLUTION THEN 2 NO. WINDOW SAMPLING / BORE HOLE UP TO 15M DEPTH TO DETERMINE SOIL PROFILE AND ALLOWABLE GROUND BEARING PRESSURE FOR MINI PILES
 - FOR RAFT FOUNDATION SOLUTION 4 NO. TRIAL PITS ADJACENT TO THE EXISTING BUILDING (LOCAL TO PRIMARY COLUMN) TO DETERMINE THE DEPTH OF THE EXISTING FOUNDATIONS (HAND DUG)
 - FOR RAFT FOUNDATION SOLUTION TAKEN FORWARD, SHEAR VANE TEST REQUIRED TO DETERMINE THE SOIL BEARING CAPACITY
 - MATERIAL SAMPLING / TESTING TO INCLUDE: ASBESTOS, WASTE ACCEPTANCE CRITERIA, CONTAMINATION SUITE
 - SOIL GAS SURVEY REQUIRED TO IDENTIFY THE POTENTIAL GAS RELEASE FROM THE EXISTING SOIL



Option 2- Framing



Option 2- Foundations and cross sections

Sustainability is considered through out the project by minimising the use the materials, sourcing material from the local suppliers within 5miles radius and steel work within 25 miles. Floor and roof construction to be timber construction .

On this project, there is an opportunity to use reused steel as some other similar corridor will be dismantled around the site.

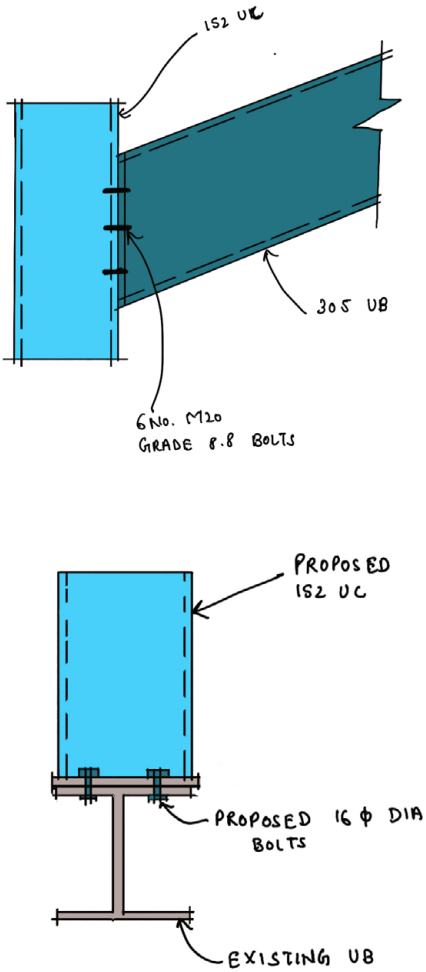
CYSTICS ROOF

Location: Manchester
Year- 2024- Ongoing
Status: Under construction

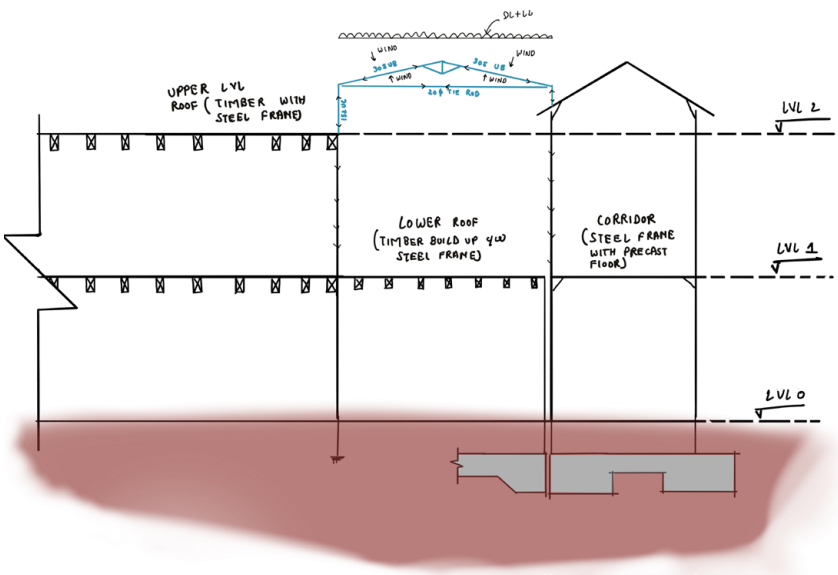
A new steel canopy was designed to cover existing M&E equipment located on the roof of an existing steel-framed building. The primary structure consists of moment and braced frames with timber floor and roof construction. The adjacent corridor structure is a steel moment frame with precast planks and a timber/steel pitched roof. The proposed canopy is supported on existing beams and columns, with lightweight cladding forming the roof enclosure. A tie rod was incorporated into the design to limit horizontal forces at the base of the structure.

My Role

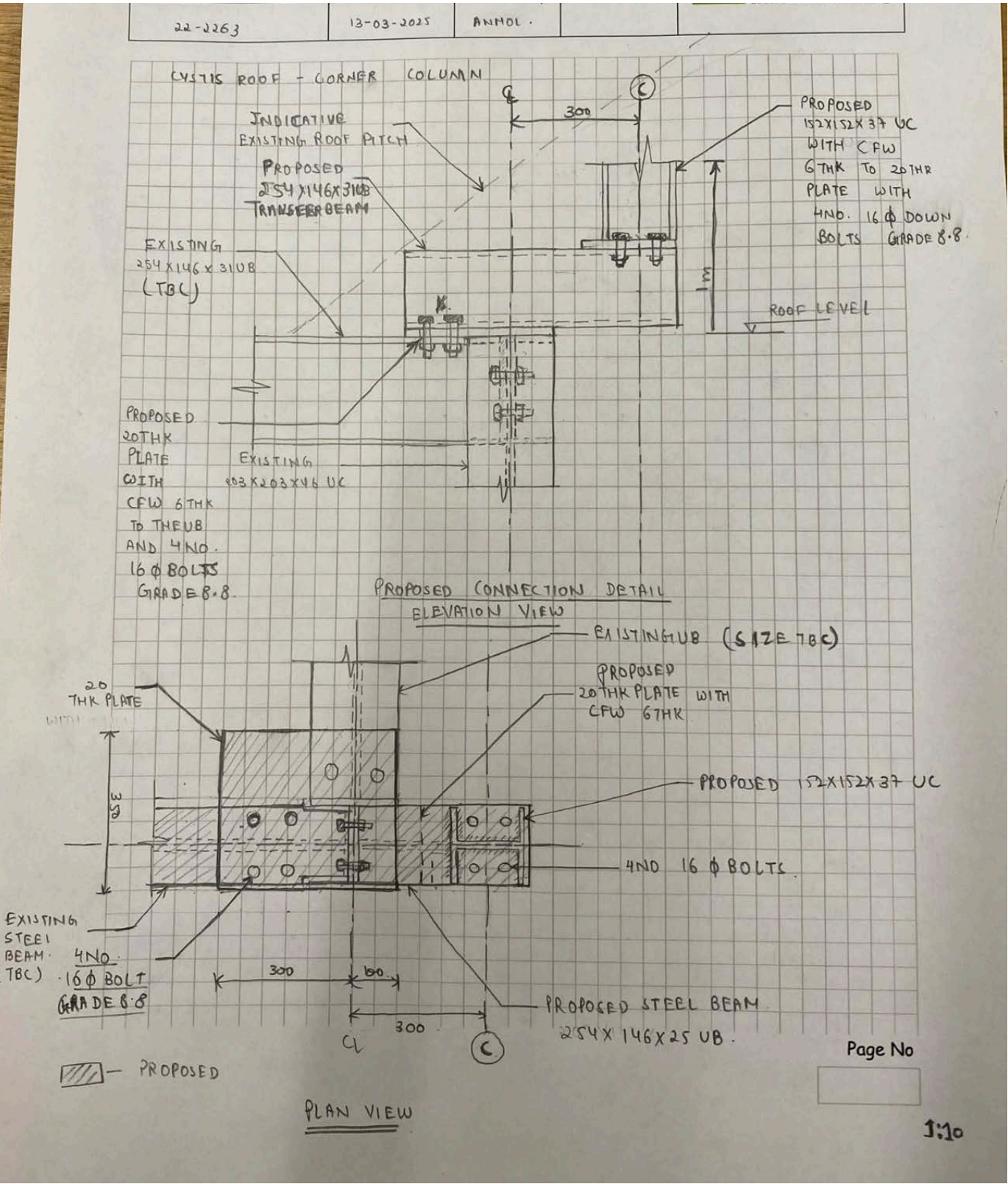
- Carried out structural surveys of the existing building.
- Modeled the steel canopy in Tekla Structures.
- Validated the model using hand calculations.
- Prepared design intents for steel connections.
- Assessed the capacity of existing beams, columns, and foundations to support the additional loads.
- Attended design team meetings and coordinated with project stakeholders.



Proposed connections



Building cross section



Initial sketching with ideas

NORTH MANCHESTER GENERAL HOSPITAL

Location: Manchester
Year- 2025
Status: Under construction

F Block Refurbishment

Refurbishment and structural upgrade of the existing F Block building to support new clinical and M&E layouts. Scope included assessment of the existing floors, coordination with architectural and services teams, and design input for local strengthening and modifications to suit new equipment and loading requirements. Reviewing the roof demolition methodology and sub contractor scaffolding design. Also advising the contractor regarding the specific concerns for loading to the existing structures.

Gantry Repair Works and Enclosures

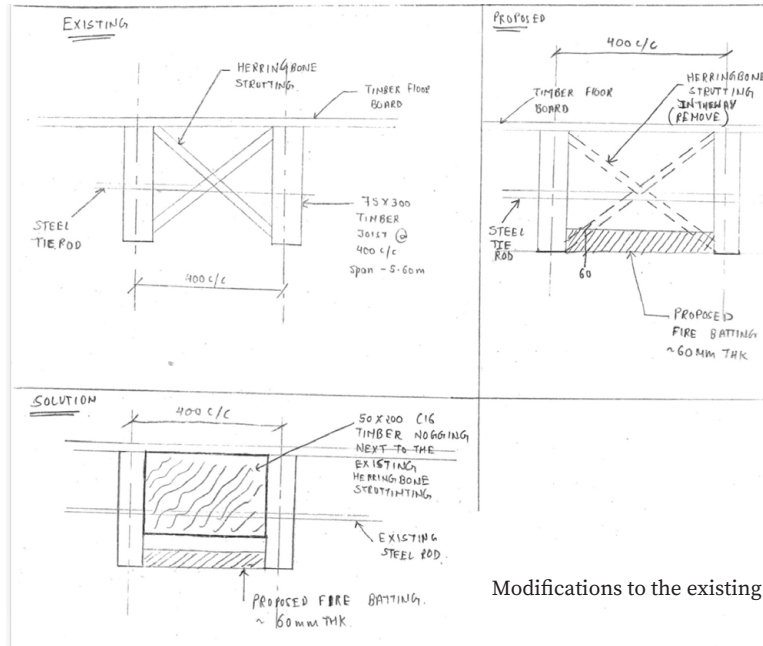
Structural assessment and repair design for existing gantry walkaway and protective enclosures. Responsibilities included site inspections, validation of as-built conditions, and preparation of design intents for repair and reinstatement, ensuring compliance with hospital safety, access constraints and specification of survey requirements.

Sub 8B D-G

Project involved construction of new foundations for existing generators, reinforced concrete slabs, switchboard bases, and associated civil works. Scope also included the design and coordination of structural supports for M&E services routed along external elevations and internal corridors, maintaining accessibility and integration with ongoing hospital operations.

RAAC Investigation and Remediation

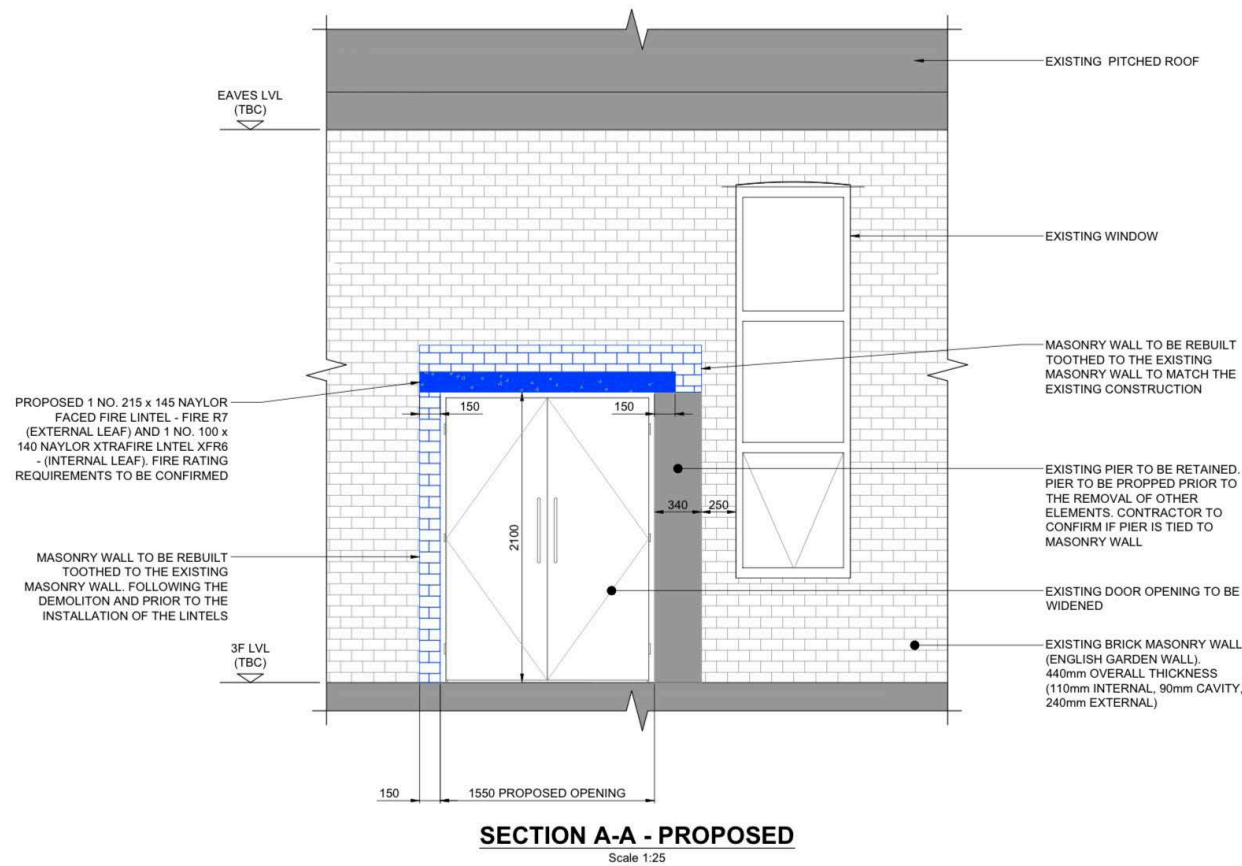
Comprehensive survey and investigation of RAAC roof planks, followed by detailed reporting, remedial design, and temporary works coordination. Responsibilities covered health and safety considerations, structural validation, and liaison with the client and design team during implementation.



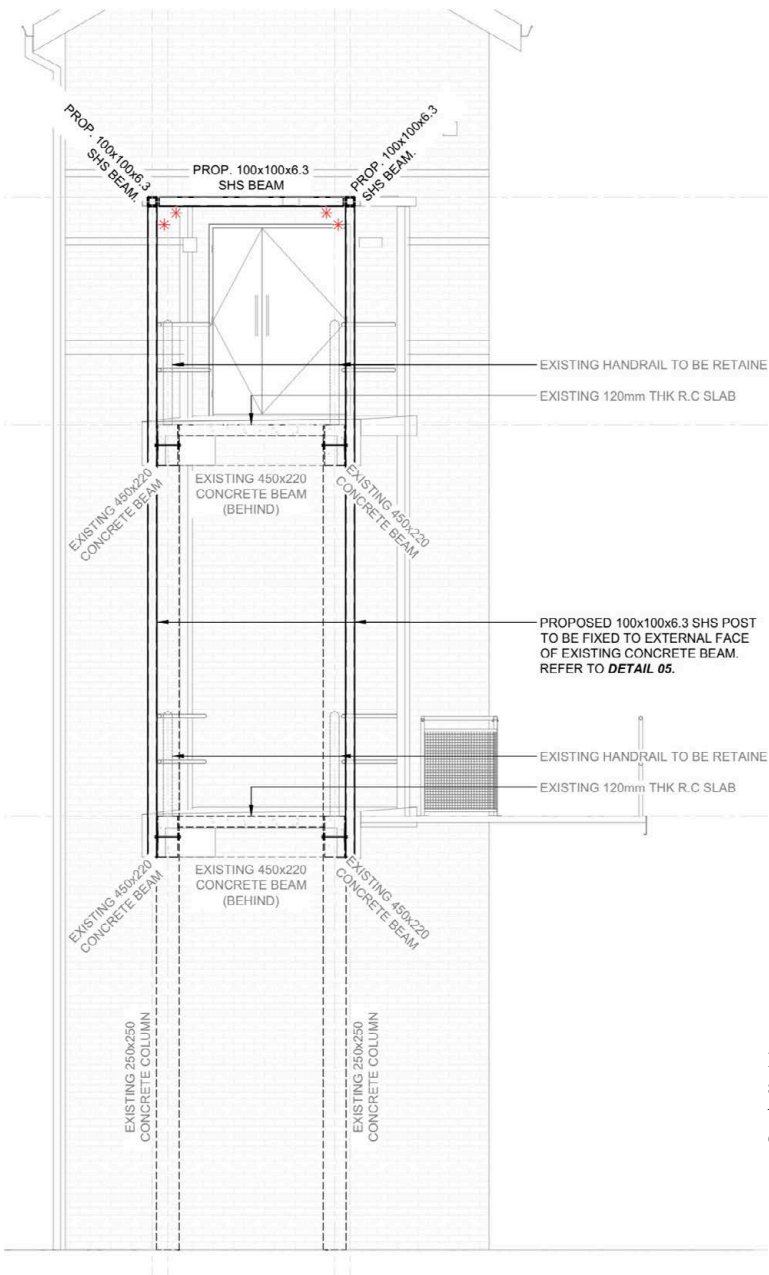
Modifications to the existing floors



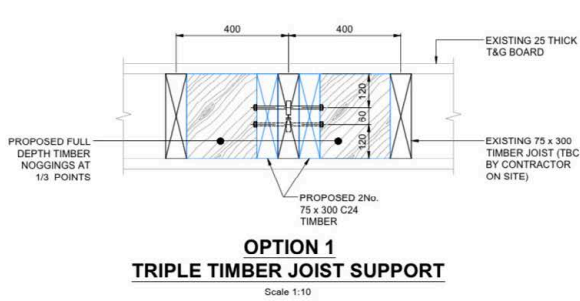
NORTH MANCHESTER GENERAL HOSPITAL



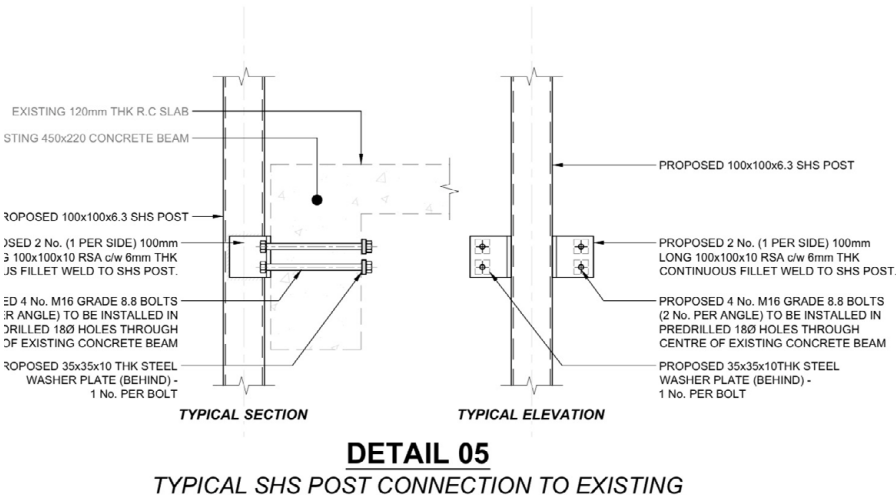
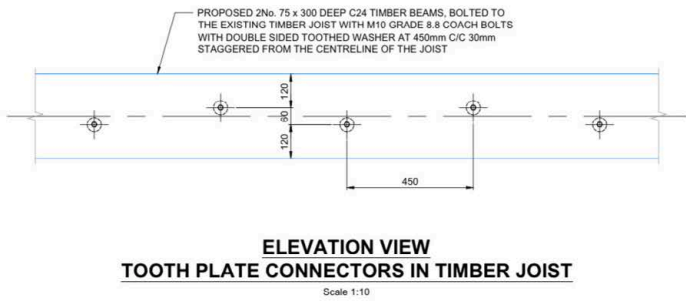
Proposed door opening detail following the site investigation through drilling and boroscope survey



Proposed enclosure to the existing concrete gantry structure. Proposed enclosure to be supported to the external face of the beam to maximize the fire escape route



Strengthening of the existing timber joist to new fire walls



Proposed steel post fixing detail

Withington CDC

Location: Manchester
Year- 2024- Ongoing
Status : Under construction

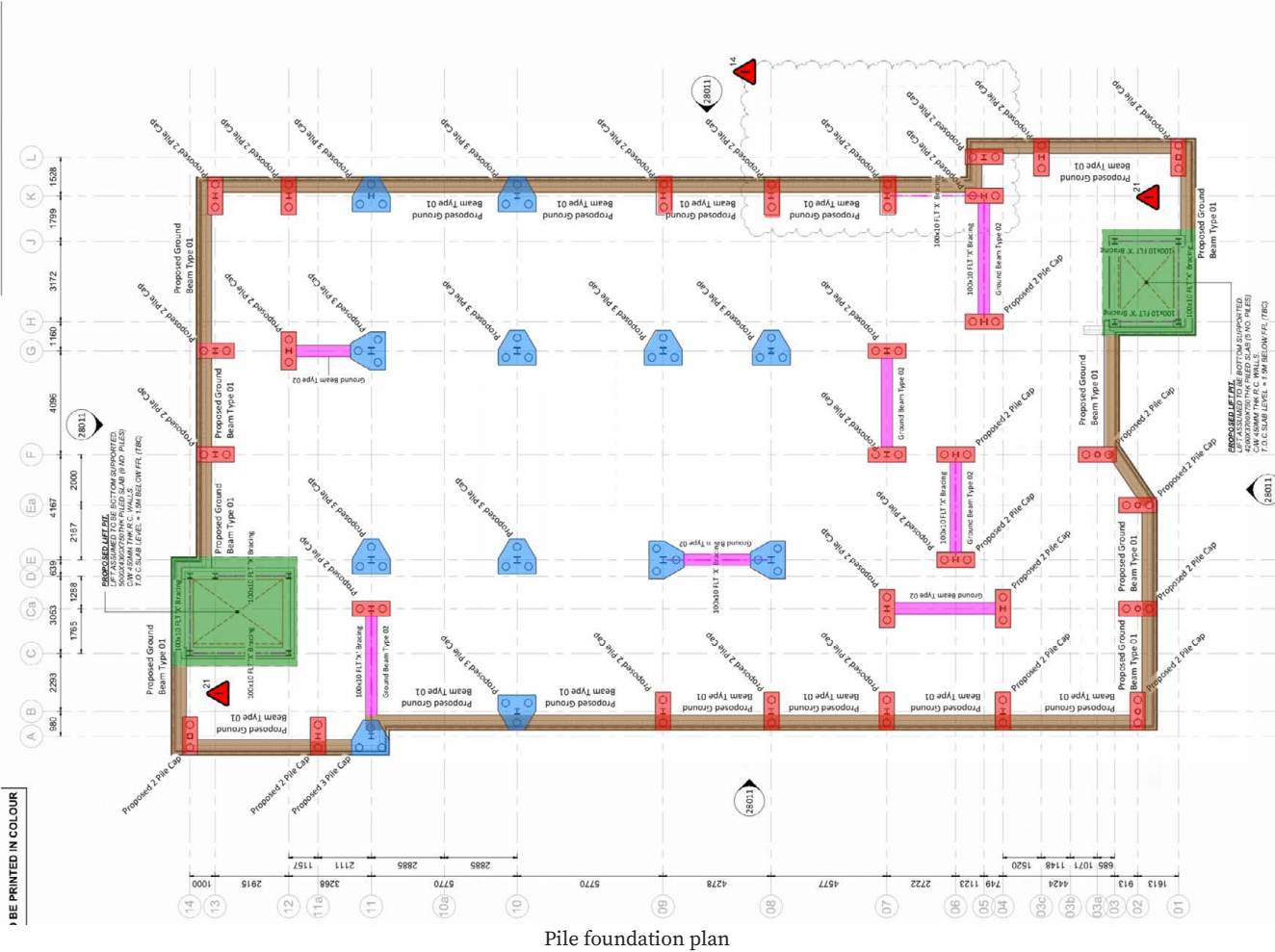
A two-storey steel frame structure supported on pile foundations, with ground-bearing slabs, composite beams with concrete slabs, and vertical bracing in both directions.

What I did as part of the team:

- Validated the structural model through hand calculations during Stage 4.
- Extracted pile loading data for Stage 5 design development.
- Involved in answering the contractor queries regarding discrepancies encountered during piling works, including pile misalignment, anchorage of piles to pile caps, and overspill issues.
- Extraction of connection forces from tekla model and checking sub contractor connection calculations.

Addition research/ works:

- Developed a Python script to post-process load data (Pile foundation and Connection forces) exported from Tekla into a standardized format, reducing manual processing time from several hours to approximately 15 minutes and ensuring results are ready for checking.
- Organised and led a site visit for the IStructE YMG North West group.



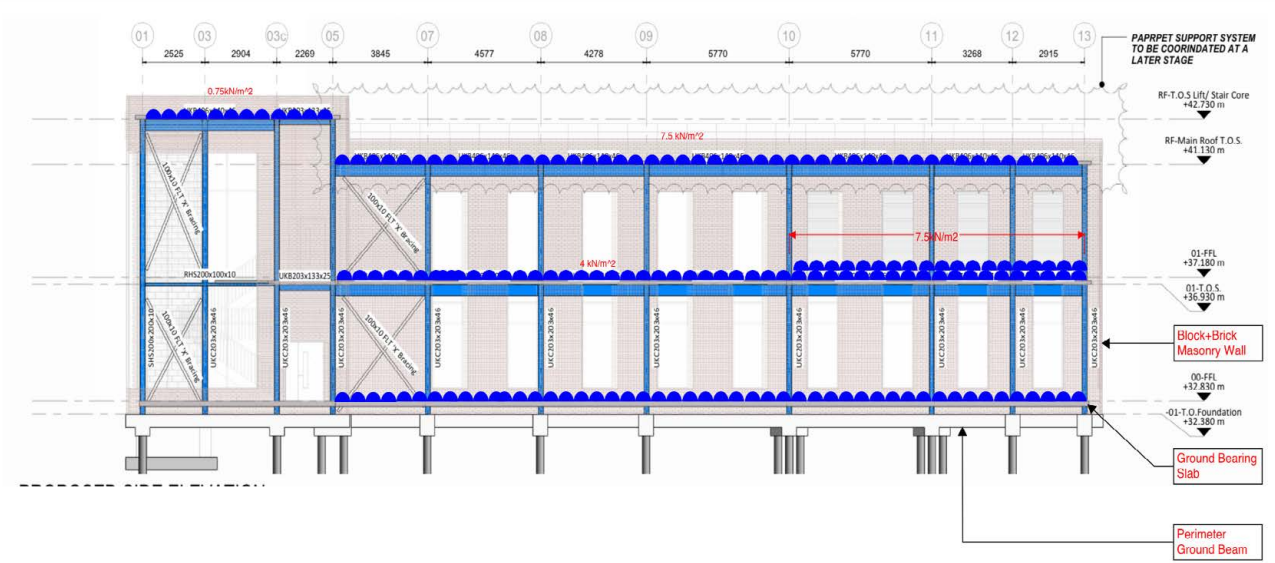
Pile foundation plan



Pile cap reinforcement cage during casting



Overall steel frame



Vertical loading diagram

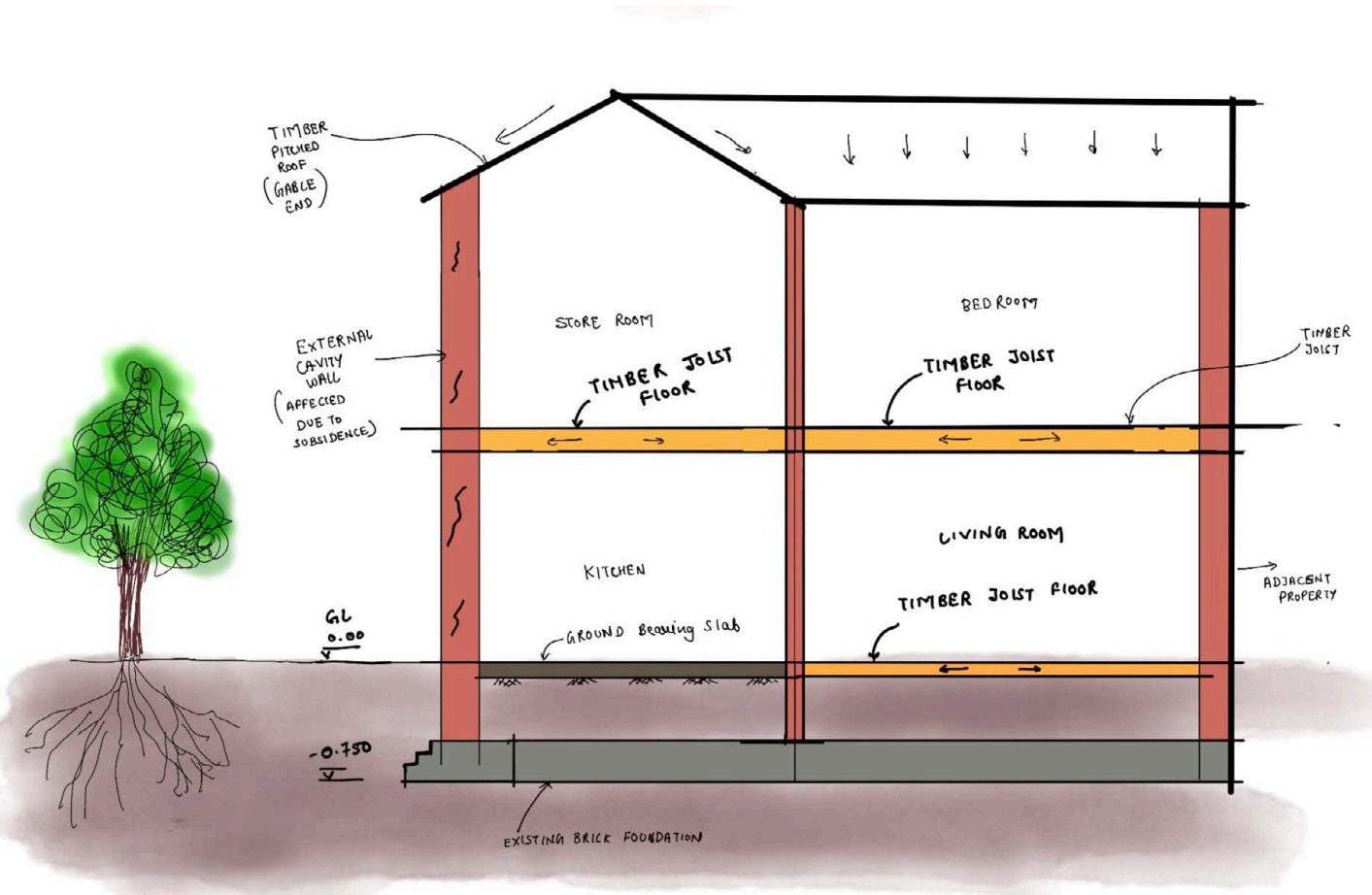
21 CHATWOOD ROAD

Location: Manchester
Year- 2025
Status:Completed

An existing brick masonry terraced house exhibiting subsidence caused by nearby trees.

My Role

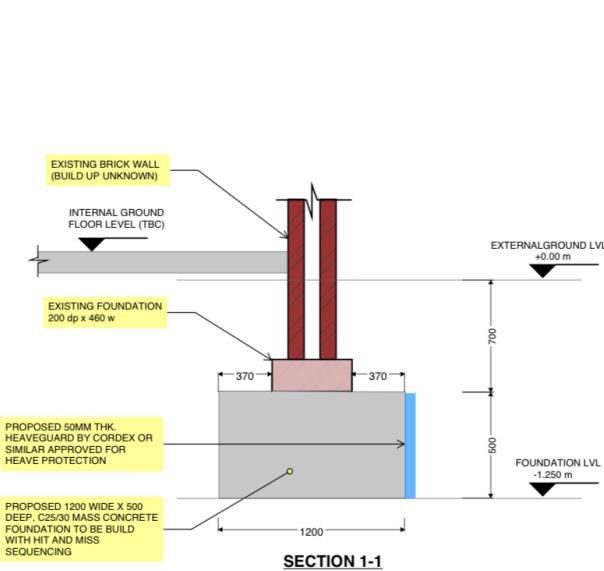
- Developed initial design intents for underpinning works, including both piled and traditional mass concrete options as requested by the client.
- The mass concrete underpinning solution was taken forward for detailed design.
- Attended design team meetings and liaised with the contractor to address technical queries during the design process
- Prepared the NBS specification document and the accompanying calculation report.
- Checked the bar bending schedule for transfer beam design option



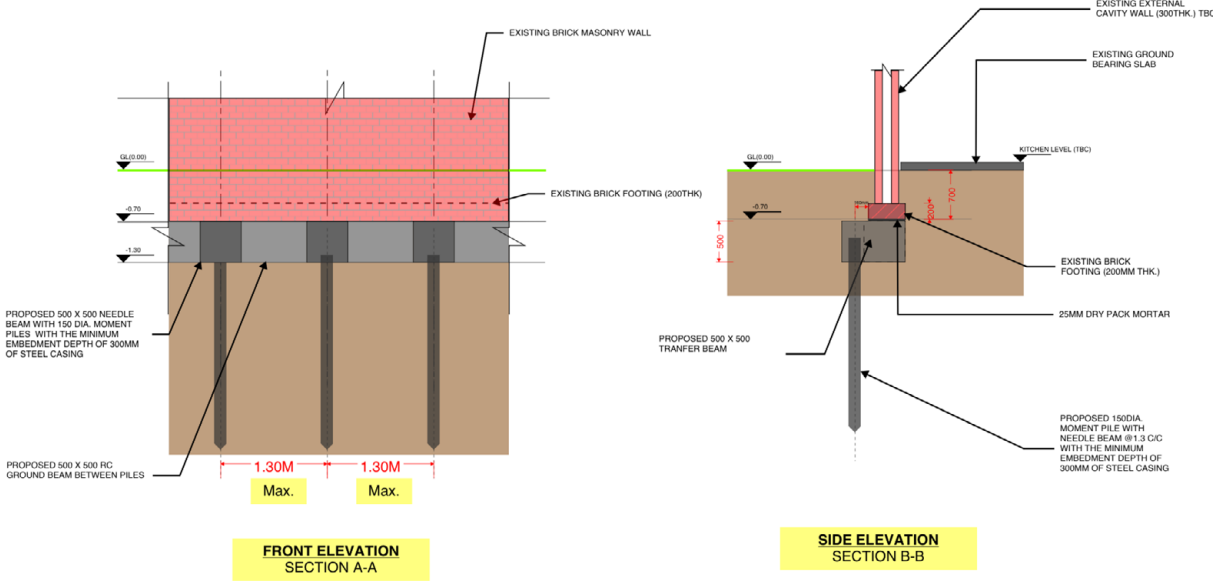
Existing structural cross section



Stepped crack in the wall



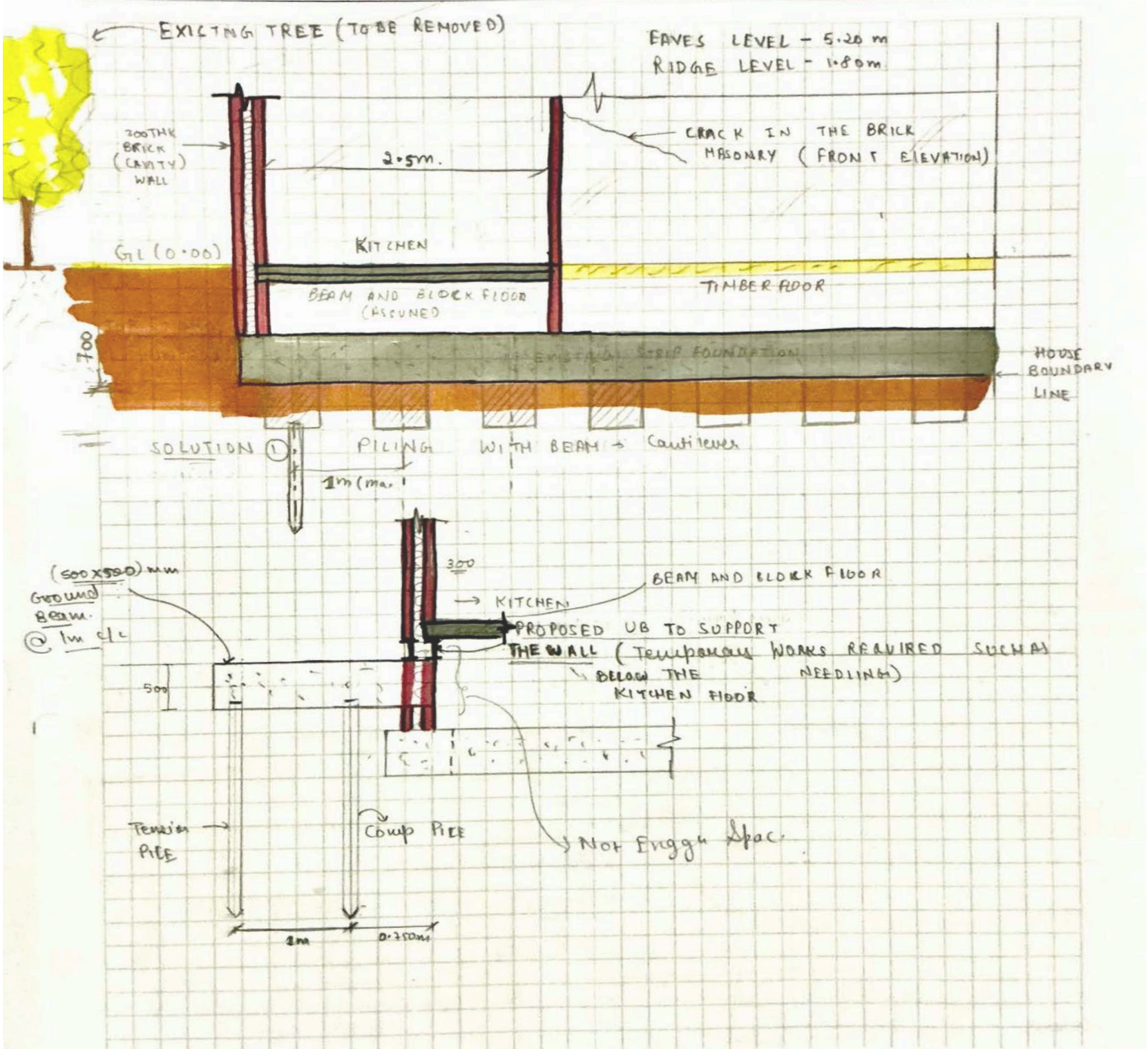
Mass concrete foundation option



Pile foundation option

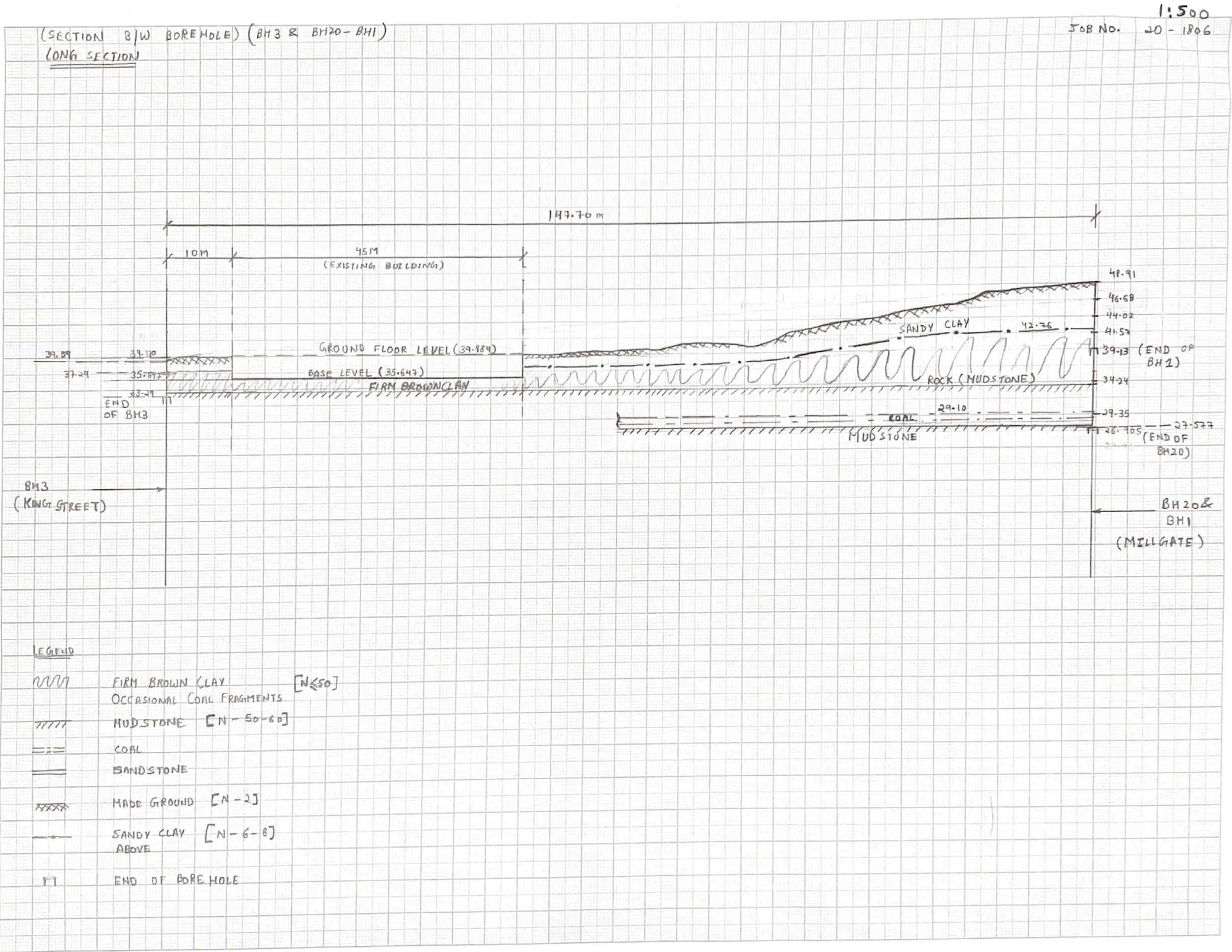
21 CHATWOOD ROAD

Initial hand sketching of the existing structure with underpinning design intent which allowed to understand the restrains with the proposals.



PRACTICE SKETCH BETWEEN EXISTING BOREHOLES

Year- 2025



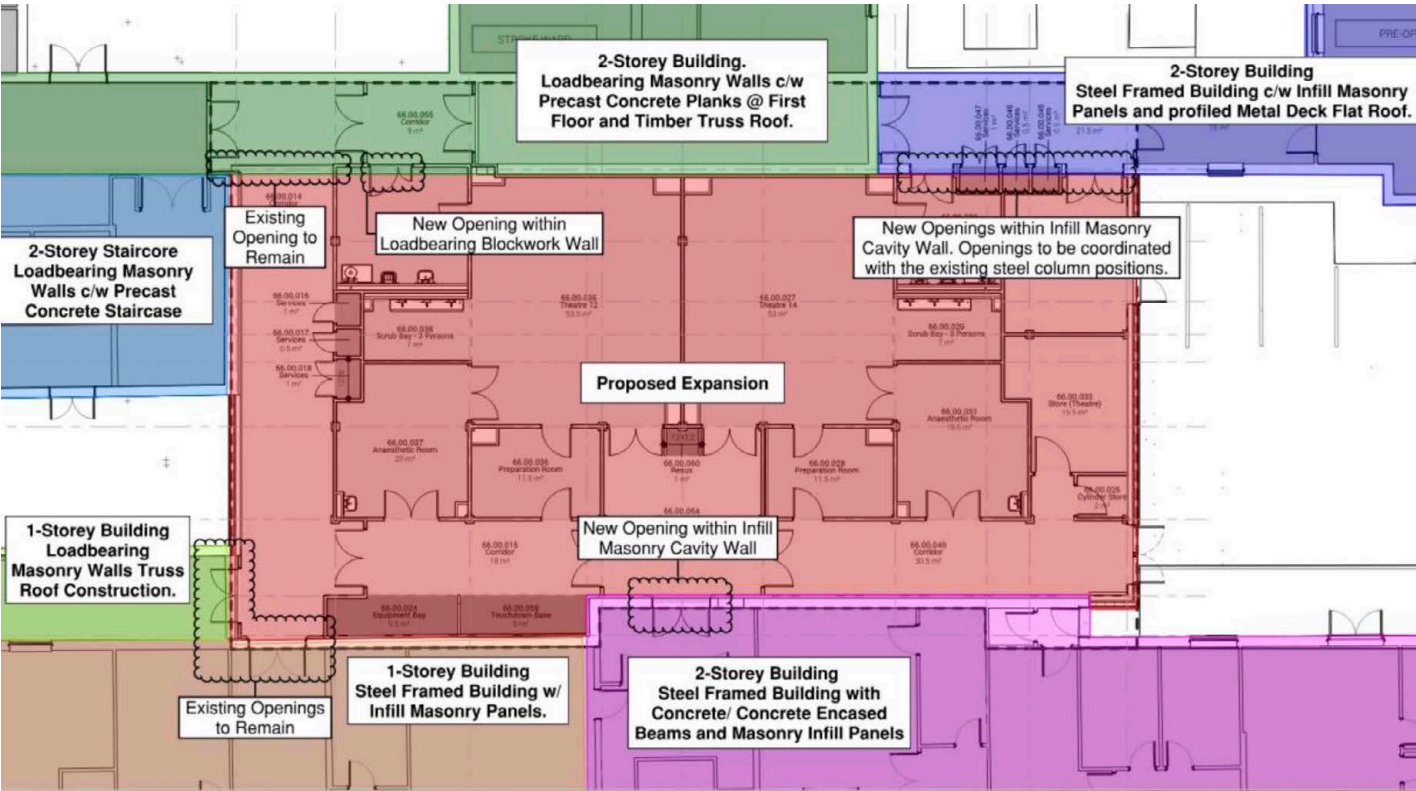
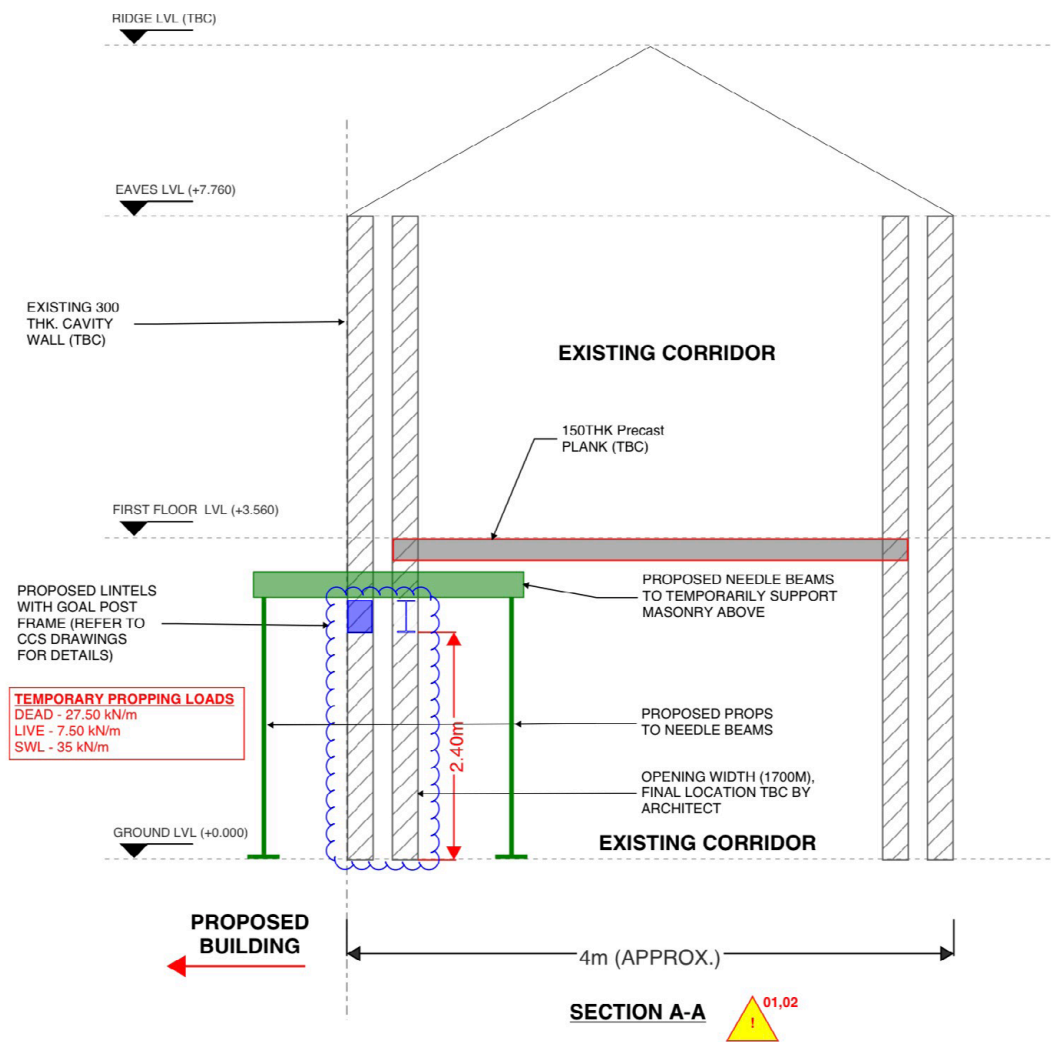
TGH 12-14 EXTENSION

Location: Manchester
Year- 2024- 2025
Status: Completed

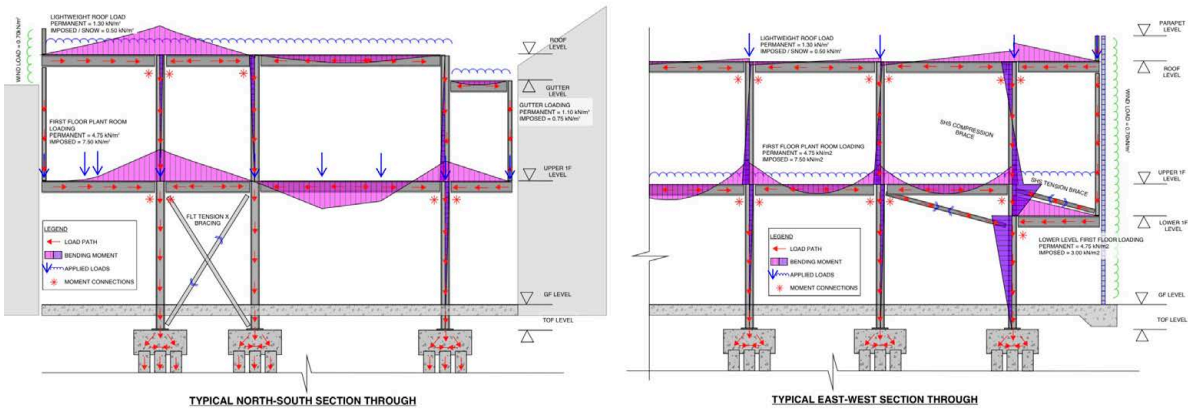
A new steel frame constructed between existing buildings, providing lateral stability through roof plan bracing, diaphragm action of the composite deck, and a combination of vertical bracing and moment-resisting sway frames. The structure is supported on pile foundations with a ground-bearing slab to minimise impact on adjacent foundations. The new frame connects to the existing buildings at both ground and first floor levels, with large ground floor openings incorporated into the design. I learned the design of the brick masonry panels, restrains, masonry pier checks.

My role:

- Designed enabling works for the construction stage in response to site findings.
- Identified the need to relocate a proposed opening to maintain the load-bearing capacity of the existing brick masonry pier and communicated the change promptly to the design team.
- Prepared design intent documentation for the contractor’s temporary works.



Overall structural strategy plan



Analysis/ Load path diagram

Bebby House

Location: University of Manchester

Year- 2022

Course: MSc Structural Engineering

Position: Student

Role: Team Leader

- In the second semester, we were tasked with designing a multi-story building as part of a six-member team. The course involved several key stages, including:
- Conceptual design Presentation to the MSc cohort
 - Conceptual design submission
 - Detailed design development and presentation
 - Final submission, incorporating a comprehensive design report and five AutoCAD drawings

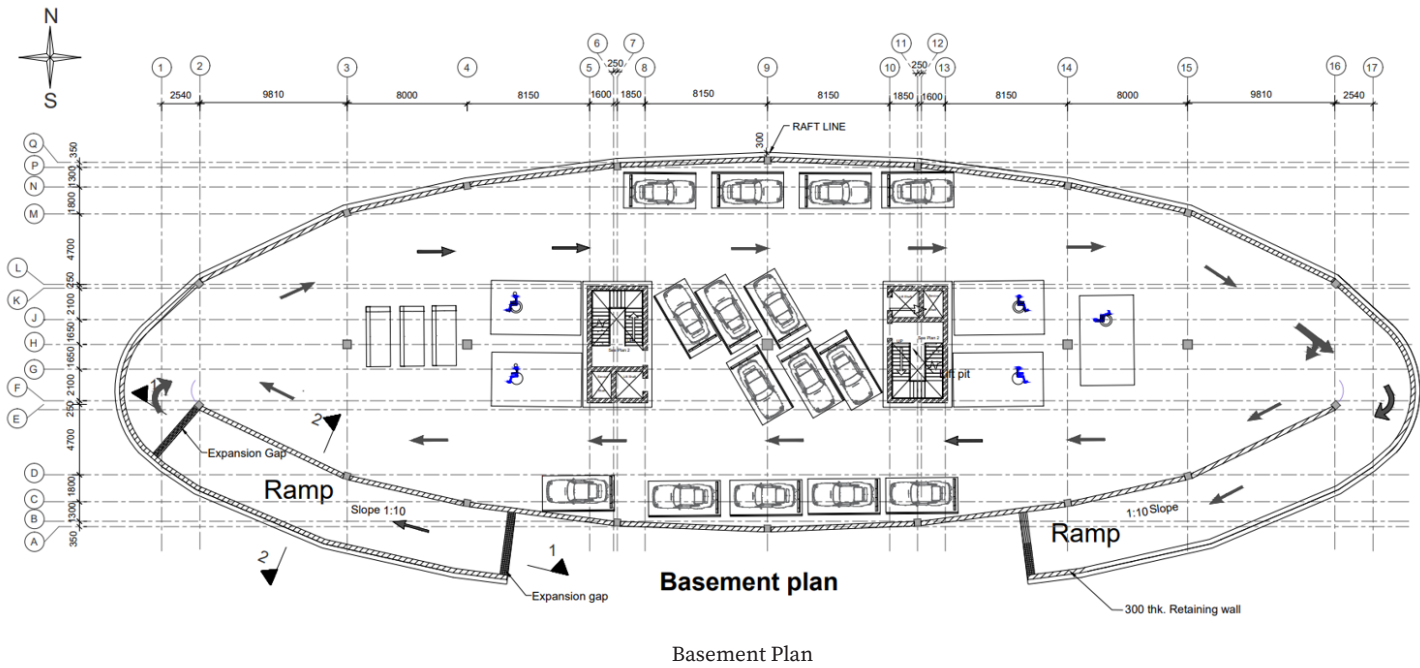
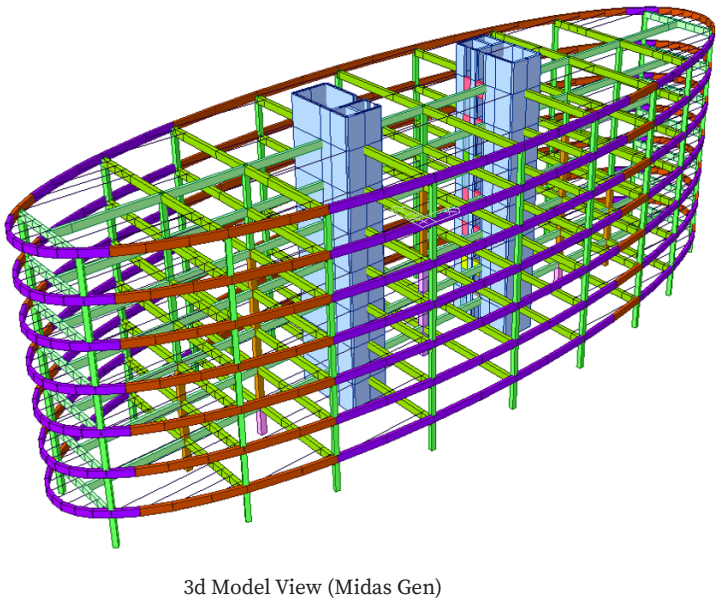
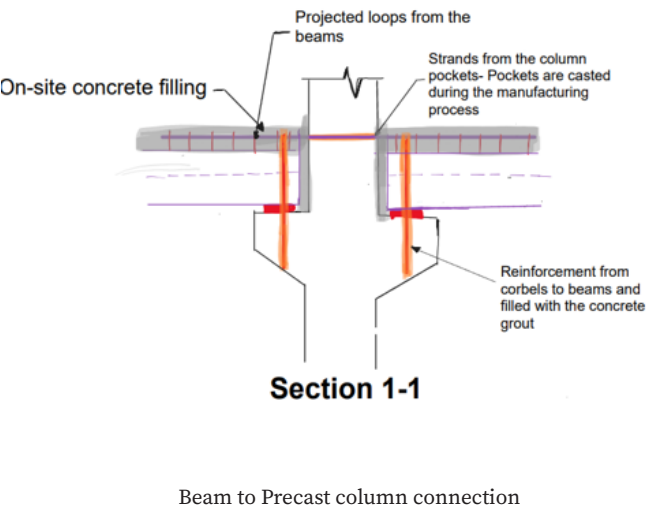
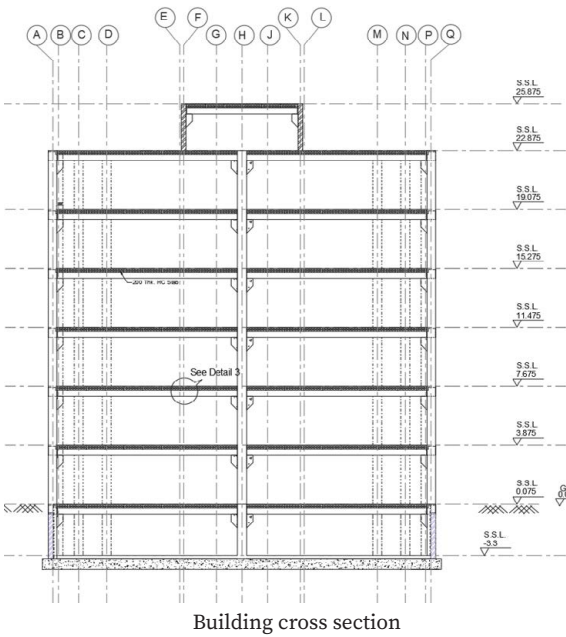
The proposed structure was an elliptical building, measuring 81m in length, 25m in width, and 6 stories high, with brownfield site location in Northwest. I led the team, facilitating effective communication and organizing regular meetings to ensure we met all submission deadlines. Additionally, I developed a conceptual design for a basement foundation using in-situ concrete and CFA piles, collaborating with a sub-team of three.

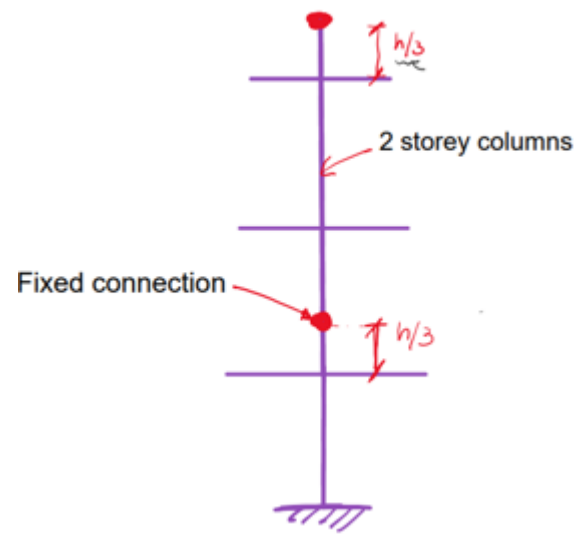
Our design incorporated a raft foundation and precast construction, aiming to expedite the build process while reducing overall costs. I took responsibility for the structural analysis using Midas Gen, overseeing the drafting work, following the design of foundations, retaining walls, shear core by doing hand calculations and ensuring the accuracy of the team’s contributions, enabling efficient workload distribution.

Our teamwork culminated in winning the “Best Structural Design” award from the university during the graduation ceremony, a testament to our collaborative success.

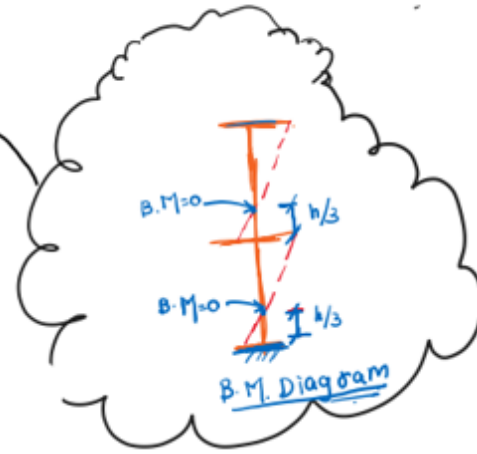
Structural Summary:

- Longest Span: 8.150m
- Raft Foundation: 800mm thick
- Precast Beams: ‘L’ and ‘T’ beams supporting 200mm thick precast hollow core slabs
- Precast Columns: Maximum size 650 x 650mm, C40/50 concrete
- Shear Cores: 300mm thick with openings

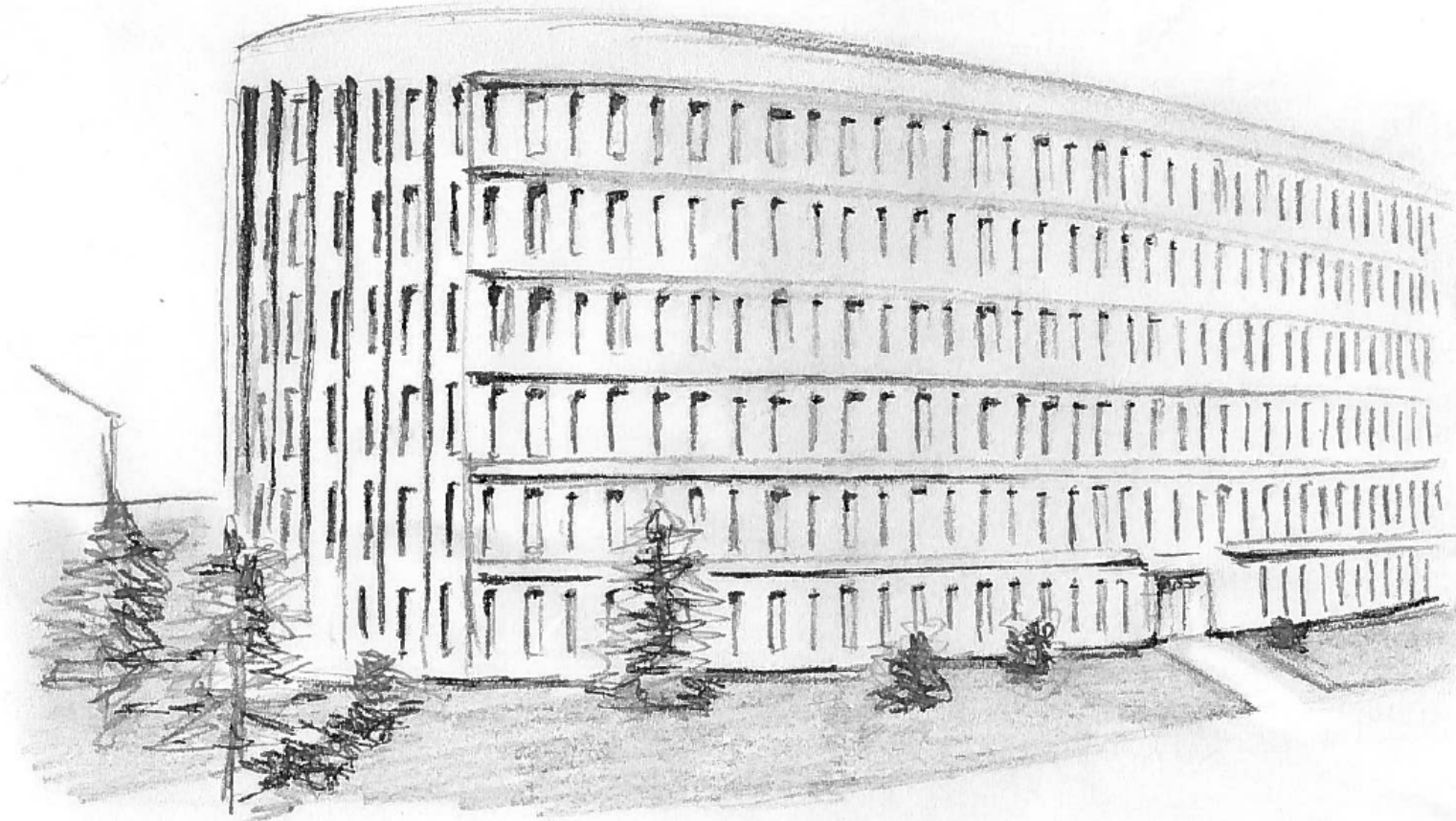
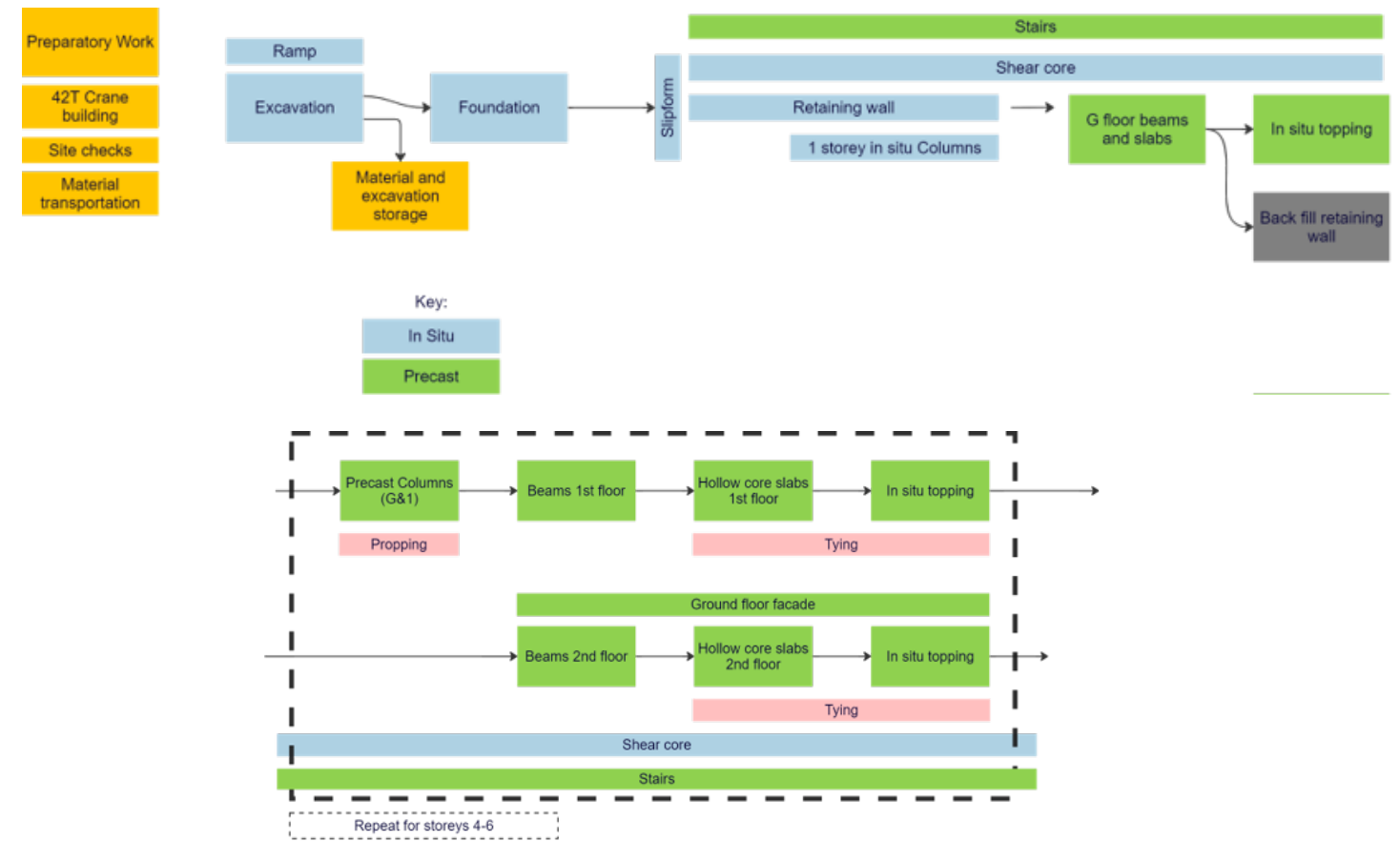




WHY?



Position of the precast column connection



Solicitare Building

Location: New Delhi
Year- 2022
Status: -

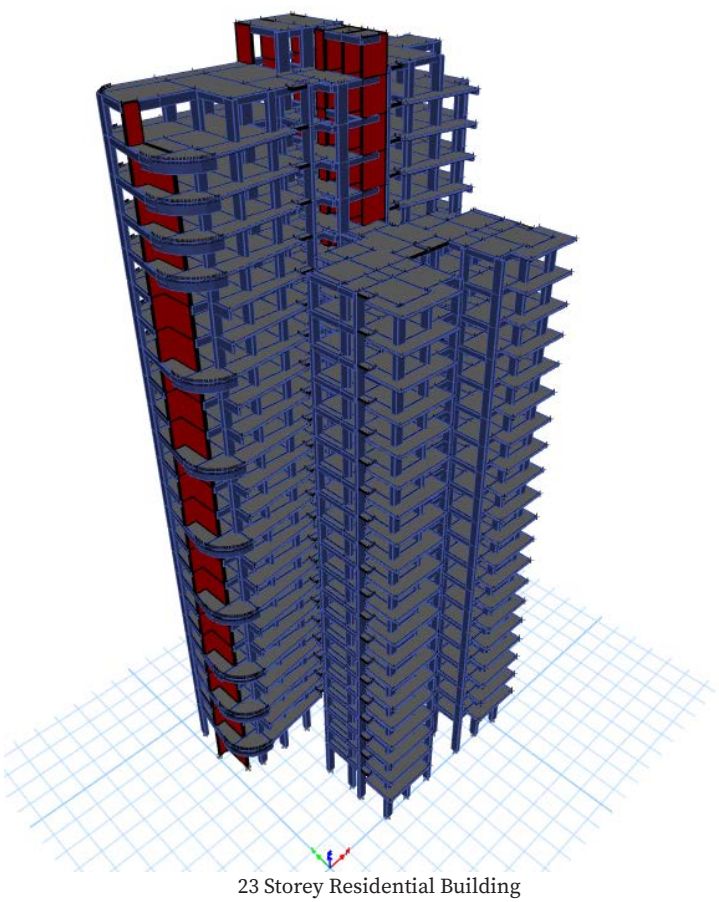
The project involved a 23-storey residential building constructed with concrete slabs, downstand beams, and columns. Shear walls were located at staircases, lift shafts, and selected other locations. Both towers are connected via a small corridor structure. The client raised concerns about excessive building movement during seismic events, with the potential for collision against a nearby building due to insufficient movement gaps.

My Role

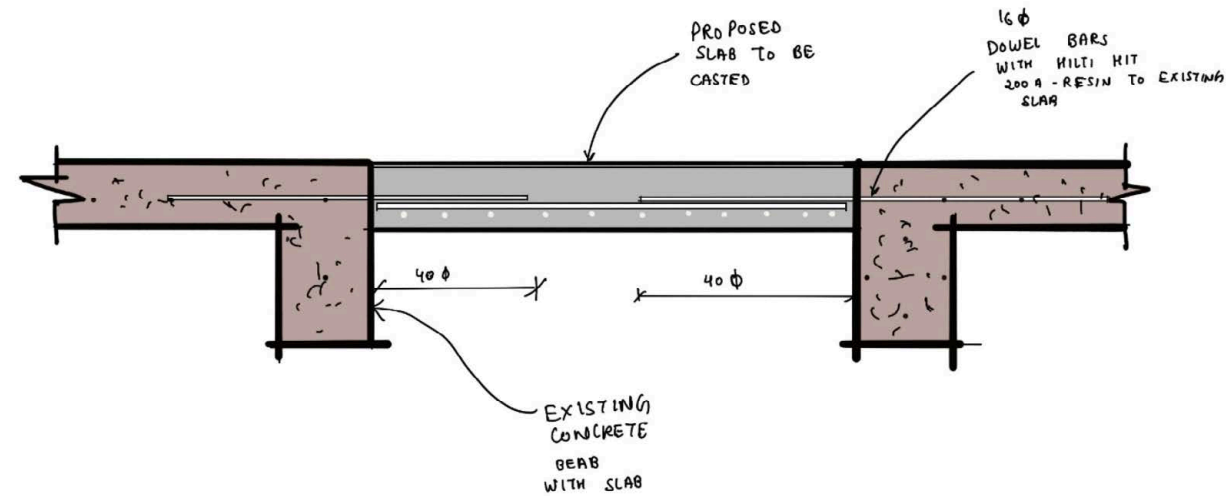
- Developed an initial model of the existing structure in ETABS.
- Performed response spectrum analysis to evaluate seismic behaviour.
- Assessed structural response across multiple vibration modes and identified critical zones.

Analysis revealed that the shear centre of the building was offset from the stronger zones, prompting the senior engineer to propose remedial works. This included constructing an additional slab adjacent to the corridor to enhance shear stiffness and reduce lateral movement.

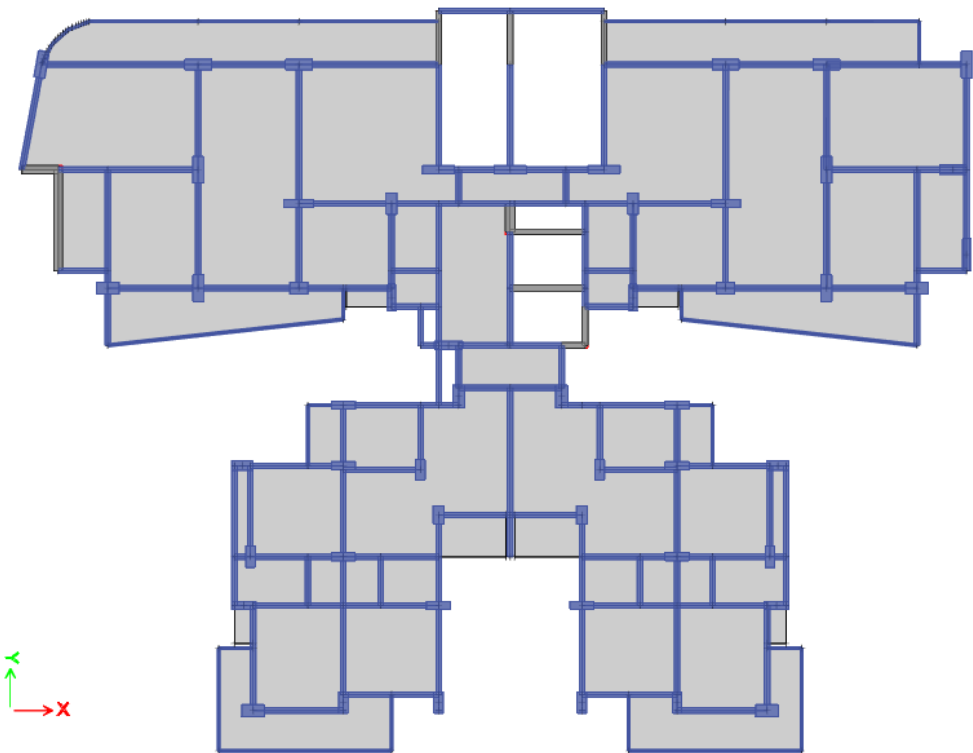
This project provided valuable experience in modeling existing structures, validating structural behaviour, and understanding the seismic performance of medium-rise buildings—insights that continue to inform my approach to structural analysis.



23 Storey Residential Building



Proposed remedials for the strengthening of the slab



Top View

IT BUILDING

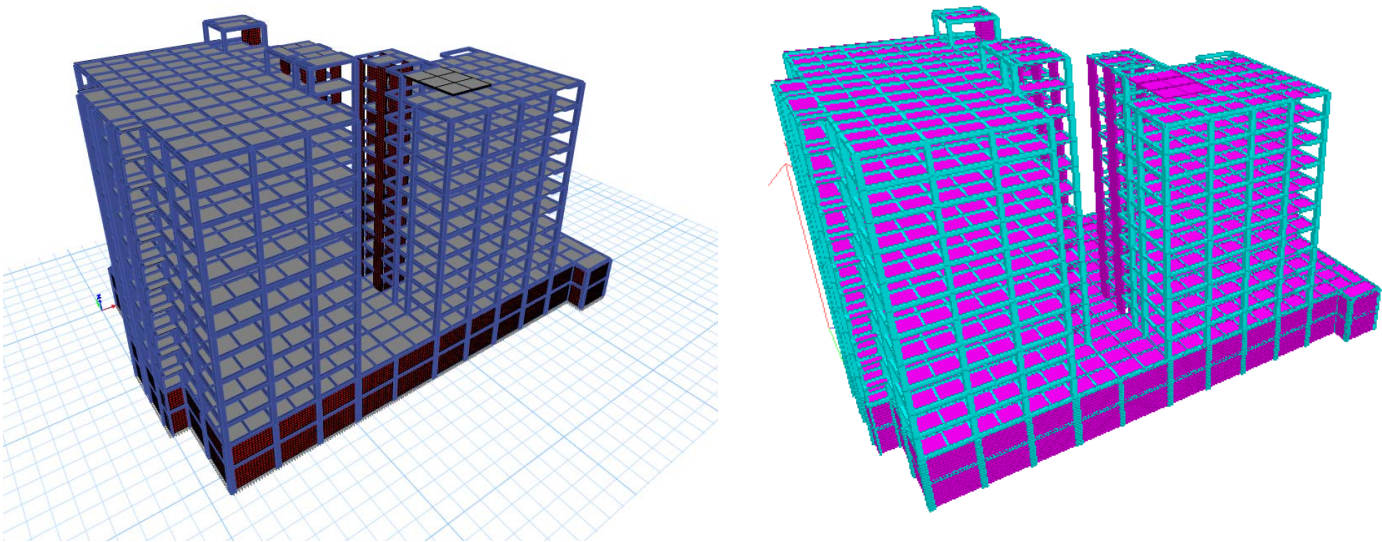
Location: Noida, India
Year- 2022
Status:-

A new 11-storey commercial building with a basement car park, designed as a traditional reinforced concrete frame with moment-resisting frames, downstand beams, two-way slabs, columns, and shear walls at stair cores. Foundations were designed as a 1,000 mm thick raft foundation. #

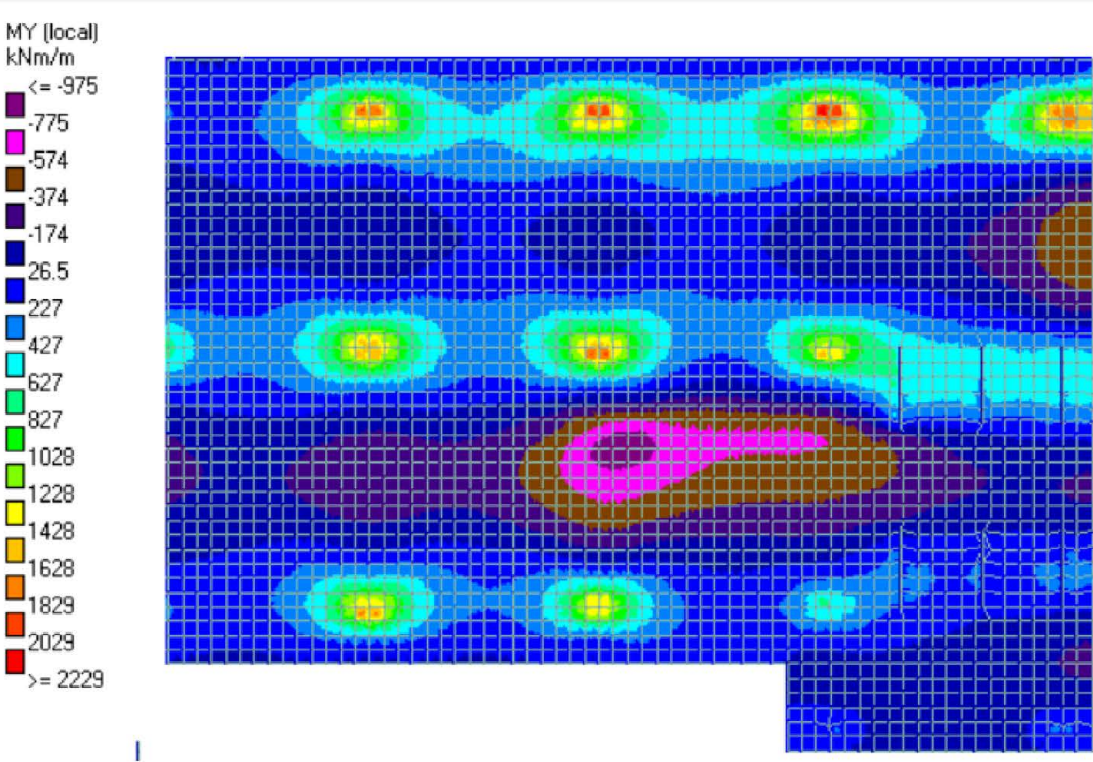
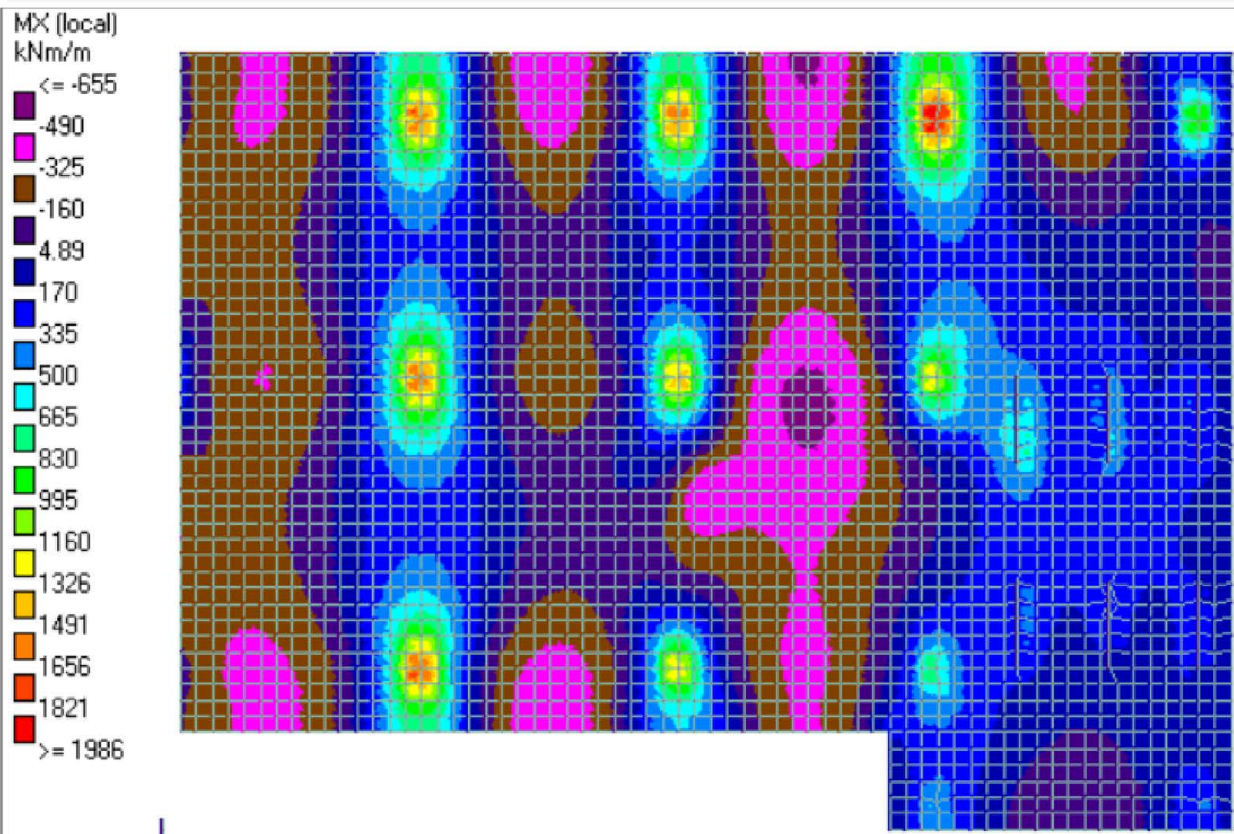
My Role

- Developed the structural model in ETABS and STAAD Pro according to the defined grid at Stage 4.
- Performed response spectrum analysis, Validated the structural model and designed key elements, including slabs, columns, beams, and retaining walls.
- Conducted finite element analysis of the raft foundation and designed reinforcement based on results and design charts.

This project provided a valuable learning experience in understanding the behaviour of raft foundations and applying finite element analysis for foundation design.



ETABS / STAAD model



Fe analysis of 1000thk. Raft foundation

ISTRUCTE YMG

